

Improper verb doubling

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Abstract: This paper presents novel data from Mandarin verb doubling to support the Williams Cycle (Williams 1974, 2003, 2011, 2013). We first demonstrate that long-distance verb doubling to a clause-medial position is only possible out of a limited class of embedding contexts, unlike its counterpart to the clausal periphery. Treating the doubling effects as consequence of VP-fronting, we propose that the varying locality profiles of verb doubling are constrained by the interaction between (i) the size of the embedding context from which the VP moves, and (ii) the height of the fronted VP's landing site in the clausal spine. We show that this constraint is captured by a strong version of the Williams Cycle, and we derive it from the Level Embedding approach developed by Williams (2003), Poole (2022), and Meadows (2024). We further show that the proposed analysis can be extended to shed new light on long-distance object shift in Mandarin Chinese.

Keywords: Verb Doubling, VP-Fronting, Locality, Williams Cycle, Mandarin Chinese

1 Introduction

It is well known that complement clauses can be transparent to some kinds of movement but not others. For example in English, finite clausal complements are transparent to VP-topicalisation (1a) but not raising-to-subject (1b).

- (1) a. **Read that novel**_i, Wassily said [_{CP} Piet did *t*_i] VP-fronting out of CP ✓
b. ***Piet**_i seems [_{CP} *t*_i read that novel]. Raising-to-subject out of CP ✗

A historically prominent approach to the clause-boundedness of movement was the *Ban on Improper Movement* (henceforth BOIM) (e.g., Chomsky 1973, 1981; May 1979), given in (2). This locality principle works in concert with an independent requirement for movement to land at certain intermediate positions (historically termed *subjacency* (Chomsky 1973), *barriers* (Chomsky 1986), or *phases* (Chomsky 2000, 2001)). The VP-movement in (1a) is permitted under the BOIM because the topicalised VP undergoes movement to matrix [Spec, CP], through the intermediate [Spec, CP], both of which are $\bar{A}$ -positions. The movement in (1b), by contrast, is restricted by the BOIM. Once movement lands at the intermediate [Spec, CP], an $\bar{A}$ -position, it cannot subsequently land at A-positions like matrix [Spec, TP].

- (2) *Ban on Improper Movement (BOIM)*
Movement cannot proceed from an $\bar{A}$ -position to an A-position. [e.g., Chomsky 1973, 1981; May 1979]

In the years since the BOIM was first proposed, the notions of A and $\bar{A}$ positions that it relies on have become more complicated: some types of movement display a mixture of A and $\bar{A}$ -properties (see early discussion in Mahajan 1990, 1994 and Webelhuth 1992, among others). It is unclear how to factor in such mixed positions into the traditional BOIM and its strict binary division of positions. More profoundly, the BOIM does not follow from any of the core ingredients of contemporary syntactic theory. Unless BOIM can be shown to follow from something fundamental (e.g., Müller and Sternefeld 1993), looking for an alternative seems desirable.

A movement constraint in a similar spirit, known as the *Williams Cycle* (Williams 1974, 2003, 2011, 2013), has been developed without reference to A/ $\bar{A}$ -positions. We give an explicit version in (3), which abstracts over a number of possible variants. In essence the Williams Cycle prevents movement lower in the clausal functional sequence (*fseq*). The BOIM and Williams Cycle are both expressions of a highly abstract *Position Intuition*: the location of the intended final landing site constrains the possible location of the base position. The unboundedness displayed by VP-topicalisation (1a) (and all archtypal $\bar{A}$ -movement), compared to raising-to-subject (1b), follows from the former landing higher in the clausal spine than the latter.

(3) *The Williams Cycle*

For certain types of movement, there is a principled connection between landing and locality such that: the higher movement lands in the clausal functional sequence, the more unbounded it is.

The goal of this paper is to offer further support for the Williams Cycle with novel data from *verb doubling* (VD) constructions in Mandarin Chinese (Li and Thompson 1981; Huang 1982; Tang 1990; Tai 1999; Paul 2002a; Cheng 2017, a.o.). We treat these as instances of VP-fronting, in which the base-position is partly realised overtly (Lai 2021, 2024; Meadows and Yan 2023, to appear). As in familiar languages like English, this movement can land in a high position in the clausal spine (4a), above where subjects typically land, namely, [Spec, TP]. We refer to this high position as [Spec, CP]. Less familiarly, such movement can also land low in the clausal spine, below the surface subject position (4b) but outside of the thematic domain (i.e., VoiceP). We refer to this lower position as [Spec, FP].¹

(4) *Two landing sites for VP-fronting with doubling in Mandarin Chinese*

- a. [CP [VP **kàn** nà-běn xiǎoshuō] [TP Piet [VoiceP **kàn** de hěn kuài]]].
 read that-CL novel Piet read DE very quickly
 ‘As for reading the novel, Piet reads it quickly.’ High VP-fronting with doubling
- b. [CP [TP Piet [FP [VP **kàn** nà-běn xiǎoshuō] [VoiceP **kàn** de hěn kuài]]].
 Piet read that-CL novel read DE very quickly
 ‘Piet reads the novel quickly.’ Low VP-fronting with doubling

Situations like this, where the same type of movement can land in multiple positions, are crucial testing grounds for the Williams Cycle. To investigate the influence of height on locality, we examine the interaction between the choice of final landing site of long-distance VP-movement, and three kinds of embedding predicate, respectively labelled ‘Group 1/2/3’ as given in (5). We provide independent evidence that these three classes of embedding predicate take complement clauses of three degrees of structural richness.² Group 1 complements are bigger than Group 2 complements, which are in turn bigger than Group 3.³ If it is correct, we expect that VP-fronting to the lower position to have a more constrained locality profile than its higher counterpart.

(5) *Three groups of embedding predicate and complement clause in Mandarin Chinese*

- a. Wassily rènwéi_{Group 1} [CP Piet **kàn** nà-běn xiǎoshuō **kàn** de hěn kuài].
 Wassily think Piet read that-CL novel read DE very quickly
 ‘Wassily thinks that Piet reads the novel quickly.’ Group 1 predicate + CP complement
- b. Wassily qiǎngpò_{Group 2} Piet [TP **kàn** nà-běn xiǎoshuō **kàn** de hěn kuài].
 Wassily force Piet read that-CL novel read DE very quickly
 ‘Wassily forces Piet to read the novel quickly.’ Group 2 predicate + TP complement
- c. Wassily chángshì_{Group 3} [FP **kàn** nà-běn xiǎoshuō **kàn** de hěn kuài].
 Wassily try read that-CL novel read DE very quickly
 ‘Wassily tries to read the novel quickly.’ Group 3 predicate + FP complement

The basic pattern of long-distance VP-movement is schematised in (6). VP-movement to the clause-initial, pre-subject position is relatively unconstrained, compared to the highly restricted VP-movement to the clause-medial, post-subject position, a phenomenon that we term *improper verb doubling*.⁴ The possible landing sites for long-distance VP-movement are contingent on the size of clausal complements containing the base position. Descriptively, it looks

¹ Abbreviations: CL = classifier, DE = postverbal modification marker, DE_{nom} = pre-nominal modification marker, NEG = negation, PFV = perfective aspect, POSS = possessive marker, PL = plural, PROG = progressive, SFP = sentence-final particle, SG = singular.

² We use CP and TP as umbrella terms for clause initial and medial fields of functional projections, in line with the spirit of Ramchand and Svenonius (2014). Clarifying the ultimate identity of these functional projections is not our main concern. TP, in particular, is suggested to unpack into several projections in Section 5, concerning the precise size of Group 2 complements and the landing site of low object shift.

³ The notation in (5b) makes clear that Group 2 complements involve object control. The restrictions we go on to show for Group 2 also hold of predicates which might involve raising-to-object such as *xiāngxìn* ‘believe’ or *qídài* ‘expect’. We do not use the latter predicates to exemplify Group 2 since their analysis remains controversial in the Mandarin literature (e.g. Ussery et al. 2016; Paul 2021). Group 3 predicates generally seem to involve subject control.

⁴ Part of the patterns we address have already been documented by Lai (2021, 2024) and Huang (2022). This work is the first to identify the full paradigm of long-distance verb doubling, including the previously unattested data concerned with what we define as Group 2 predicates.

like VP-movement to the clause-medial position cannot cross Group 1 or Group 2 clause boundaries. This is exactly what the Williams Cycle leads us to expect if VP-fronting can in principle land below TP.

(6) *Schematic long-distance VP-movement*

		Landing sites
a.	(VP) Subj (*VP) Pred. _{Group 1} [CP ... VP ...]	✓ Subj ✗
b.	(VP) Subj (*VP) Pred. _{Group 2} [TP ... VP ...]	✓ Subj ✗
c.	(VP) Subj (VP) Pred. _{Group 3} [FP ... VP ...]	✓ Subj ✓

The Williams Cycle (as given in (3)) is a descriptive constraint on movement which, if it is to improve upon the BOIM, also needs to follow from general properties governing the narrow syntax. We outline a version of the *Level Embedding* approach pioneered by Williams (2003, 2011, 2013) and more recently developed by Poole (2022) and Meadows (2024). The central innovation of Level Embedding, *Parallel Derivation* (7a), is rethinking the timing of main and embedded clause structure building. The result of (7a) is that the bigger a clause is, the later in the derivation it is embedded - with embedding achieved through *Late Merge* (7b). The *Strict Cycle Condition* (7c) interacts with this novel timing of structure building and embedding to restrict long-distance movement. Long-distance movement can only be triggered by features introduced at stage of the derivation when embedding takes place, or later.

(7) *Three key ingredients of Level Embedding*

- PARALLEL DERIVATION:** The same component of the clausal *fseq* (V, Voice, F, T, C, etc.) is externally merged in every workspace at the same time.
- LATE MERGE:** Clausal embedding involve external Late Merge.
- STRICT CYCLE CONDITION:** The output of prior cycles of rule application cannot be altered by rules in subsequent cycles. Every functional projection in the clausal *fseq* delineates a new cycle.

An analysis of improper verb doubling in terms of A/ $\bar{A}$ -distinction is conceivable: low verb doubling could be archetypal A-movement, which is not able to skip over intervening VPs. Setting aside what such an approach would need to say about Group 3 predicates, its key predictive difference with Level Embedding is that the latter expects low nominal movement, analogous to low verb doubling, to be similarly restricted. This expectation is borne out, by close attention to long-distance *object shift* (OS) (Ernst and Wang 1995; Shyu 1995; Paul 2002b, 2005; Chen 2023, to appear, a.o.). This nominal movement can likewise land in a pre-subject or post-subject position (8), but targeting [Spec, TopP] in the clause-medial domain - a position slightly higher than the [Spec, FP] landing site of low verb doubling.

(8) *Two landing sites for object shift in Mandarin Chinese (within a rich clause structure)*

- [CP [DP **Nà-běn xiǎoshuō**] [TP Piet [_{VoiceP} kàn ___ de hěn kuài]]].
that-CL novel Piet read DE very quickly
'THAT NOVEL, Piet reads quickly.'
High object shift
- [CP [TP Piet [_{TopP} [DP **Nà-běn xiǎoshuō**] [_{VoiceP} kàn ___ de hěn kuài]]]].
Piet that-CL novel read DE very quickly
'Piet reads THAT NOVEL quickly.'
Low object shift

What emerges from comparing verb doubling and object shift are three locality profiles for embedding contexts (9). Group 1 complements let high VD/OS escape, but trap low VD/OS. Group 2 complements let high VD/OS and low OS escape, but trap low VD. Group 3 are the least restrictive, allowing high and low VD/OS to escape. We cannot treat the boundedness of low OS as a reflex of archetypal A-movement: low OS can move over intervening objects of Group 2 predicates. What we are seeing is precisely what the Level Embedding/Williams Cycle expect: the bigger the complement clause is, the higher we have to aim to land in the clausal *fseq* in order to escape the complement.

(9) *Three locality profiles for embedding contexts*

Locality profile	Pred. _{Group 1} + [CP ...]	Pred. _{Group 2} + [TP ...]	Pred. _{Group 3} + [FP ...]
Transparent to:	High VD/OS	High VD/OS, Low OS	High VD/OS, Low VD/OS
Opaque to:	Low VD/OS	Low VD	∅

The paper is organised as follows. Sect. 2 outlines several basic properties of verb doubling in Mandarin Chinese

that favour a VP-fronting analysis. Sect. 3 presents the paradigm of long-distance VP-fronting in different embedding contexts, and argues for the idea that embedded clausal complements vary in size. Sect. 4 sketches out the Williams Cycle in the first place, and then explores the Level Embedding approach in greater detail. Sect. 5 respectively extends the Level Embedding account to restrictions on long-distance object shift. Sect. 6 concludes the paper by reviewing some theoretical issues arising from improper verb doubling and our account of it.

2 Background on Mandarin verb doubling

In this section we provide some background on verb doubling constructions in Mandarin Chinese, and argue for the idea that such structures are derived from VP-fronting operations that target the specifier position of two different functional projections in the clausal spine.

2.1 Basic data: high and low verb doubling

Verb doubling in Mandarin Chinese has been well-documented in the literature (e.g., Li and Thompson 1981; Huang 1982; Tang 1990; Tai 1999; Paul 2002a; Cheng 2017). We refer to the double verbs as a consequence of displacement, and, based on the position of the first verb with the object relative to the surface subject, classify verb doubling constructions into two sub-types, as exemplified in (10).

- (10) a. Piet **kàn** nà-běn xiǎoshuō hěn kuài.
Piet read that-CL novel very quickly
'Piet reads the novel quickly.' No verb doubling
- b. [**Kàn** nà-běn xiǎoshuō] Piet **kàn** de hěn kuài.
read that-CL novel Piet read DE very quickly
'As for reading the novel, Piet reads it quickly.' High verb doubling
- c. Piet [**kàn** nà-běn xiǎoshuō] **kàn** de hěn kuài.
Piet read that-CL novel read DE very quickly
'Piet reads the novel quickly.' Low verb doubling

(10a) demonstrates a canonical structure with two postverbal elements in Mandarin Chinese (cf. Huang 1982): the lexical verb *kàn* 'read' is followed by the direct object *nà-běn xiǎoshuō* 'read the novel' and the manner adverbial *hěn kuài* 'very quickly' at the same time.⁵ In its verb doubling counterparts, the first verb, together with the object, shows up either in a pre-subject/clause-initial position (10b) or a post-subject/clause-medial one (10c). We term (10b) and (10c) respectively *high* and *low* verb doubling. In both, the verb is doubled and also pronounced in its base position before the adverbial and the particle *de*.

Interpretation-wise, (10b) clearly involves either topicalisation or focalisation (as suggested by its translation); by contrast, the meaning of (10c) is not sufficiently distinct from that of (10a). Both low verb doubling and the structure without doubling are informationally neutral and interchangeable in most situations, as illustrated in the following scenario (11).

- (11) *Scenario: One day Pablo walked into the classroom and saw Piet reading Three-Body Problem very quickly. Later Pablo mentioned this to Wassily ...*
- a. Piet **kàn** Sāntǐ hěn kuài.
Piet read TBP very quickly
'Piet was reading the *Three-Body Problem* (TBP) quickly.' No VD: informationally neutral
- b. Piet **kàn** Sāntǐ **kàn** de hěn kuài.
Piet read TBP read DE very quickly
'Piet was reading the *Three-Body Problem* quickly.' Low VD: informationally neutral
- c. #**Kàn** Sāntǐ Piet **kàn** de hěn kuài.
read TBP Piet read DE very quickly

⁵ See Lai (2021, 2024) for discussion about relationship between postverbal materials and verb doubling.

‘As for reading the *Three-Body Problem*, Piet was reading it quickly.’ High VD: informationally marked

Note that larger chunks of thematic structure including applied arguments can also participate in the construction of verb doubling (H.-L. Huang 2018; Meadows and Yan 2023), as shown by the double object constructions in (12). Since constructions like these demonstrate the same results in various cross-clausal contexts to be discussed in Sect. 3.1, we do not include them in the rest of the paper.

- (12) a. [**Jiào** Piet yǔyánxué] Wassily **jiào** de hěn kuài.
 teach Piet linguistics Wassily teach DE very quickly.
 ‘As for teaching Piet linguistics, Wassily teaches him quickly.’ High VD with double objects
- b. Wassily [**jiào** Piet yǔyánxué] **jiào** de hěn kuài.
 Wassily teach Piet linguistics teach DE very quickly.
 ‘Wassily teaches Piet linguistics quickly.’ Low VD with double objects

It is also worth noting that all verb doubling constructions seen so far contain the particle *de*. There are other particles possible in verb doubling constructions, for instance, aspect markers *le* and *guò* (as discussed concretely in Lai 2021, 2024). Those particles are compatible with other postverbal adverbial constituents, resulting in verbal modification with different meanings (see more details in Li and Thompson 1981; Tai 1999; Cheng 2007, 2017). For simplicity, we illustrate the patterns to be discussed in Sect. 3.1 only with the ‘manner *de*’, but the full paradigm holds across all sorts of verb doubling constructions.

2.2 A phrasal movement approach to verb doubling

We treat verb doubling as movement of the lower thematic domain, minimally containing the direct object, which we refer to as VP for simplicity (see the same line of thought in Lai 2021, 2024 and Meadows and Yan 2023, to appear).⁶ As observed in the previous section, the fronted VP can take a position either following or preceding the subject. Following the idea that the surface subject in Mandarin Chinese typically occupies [Spec, TP] (Lin 2011), we propose that there are two possible landing sites for the VP, namely, a relatively lower position in the clausal spine [Spec, FP], and a higher one [Spec, CP]. Given that low verb doubling is informationally neutral compared to high verb doubling (recall the contrast in (11)), we postulate that the former is triggered by the categorial feature [*uV*] located in the medial clausal spine for producing one kind of unmarked word order (13), and that the latter is triggered by the feature [*uTop*] in the left periphery for the sake of information structure (14).



Evidence in favour of a phrasal movement analysis includes a language-specific diagnostic, i.e., *lexical identity effects*, and the conventional movement tests, namely, *island effects* and *reconstruction effects*. We start off with the lexical identity effects in what follows. In Mandarin Chinese, it is possible for two words to be interchangeable in many occasions, if they share the same lexical meaning but not the same form.⁷ As exemplified in (15), while *kàn* ‘read’ is used in the question, the answer employs *dù*, a different word form with (nearly) the same meaning ‘read’, and is still considered a licit response.

⁶ There are a variety of other movement approaches to verb doubling, for instance, Gouguet (2006), Cheng (2007), Yu (2017), H.-L. Huang (2018), Bartos (2019), Yin (2022), etc.

⁷ Faruk Akkuş (pers.comm.) wonders whether there exists a subset-superset relation among the lexically-identical words, and whether such a relation would affect verb doubling. As far as we can tell, at least with respect to the examples concerned here, it is not clear if there exists such a relationship between *kàn* and *dù*, but swapping the positions of these two words would not change the results given in (15) and (16).

(15) *Interchangeable lexical items*

- Q: Nǐ zài **kàn** shénme ya?
2SG PROG read what SFP
'What are you reading?'
A: Wǒ zài **dú** Sāntǐ ne.
1SG PROG read TBP SFP
'I am reading the *Three-Body Problem*.'

However, verb doubling is one of those occasions in which the interchange of words with same meaning is prohibited, as given in (16). In either high (16a) or low verb doubling (16b), *kàn* and *dú* are unable to form a grammatical doubling construction. Instead, doubled verbs have to be identical (that is, *kàn* occurs twice in the structure), leading to the lexical identity effects.

(16) *Lexical identity effects in verb doubling*

- a. **Kàn** nà-běn xiǎoshuō Piet **kàn**/***dú** de hěn kuài.
read that-CL novel Piet read DE very quickly
'As for reading the novel, Piet reads it quickly.'
b. Piet **kàn** nà-běn xiǎoshuō **kàn**/***dú** de hěn kuài.
Piet read that-CL novel read DE very quickly
'Piet reads the novel quickly.'

Under the copy theory of movement (e.g., Chomsky, 1993, 1995; Nunes, 1995), such effects can receive a direct explanation, namely, both verbs are actually VP-copies, and one of them undergoes phrasal movement with the direct object to some higher position in the clausal spine. In other words, the higher VP-copy gets spelled out completely. We will come back to the spell-out of the lower VP-copy by the end of the section.

Further evidence for a VP-movement analysis for verb doubling comes from island and reconstruction effects. In (17a) and (18a), we construct two canonical structures without verb doubling, and put the VP *kàn xiǎoshuō* 'read novels' in a *wh*-island and a complex NP-island respectively. In both (17b) and (18b), high verb doubling out of islands results in ungrammatical sentences, which is expected if verb doubling is VP-movement.

(17) *Wh-islands for high verb doubling*

- a. Wassily xiǎng zhīdào [wèishéme Piet **kàn** xiǎoshuō hěn kuài].
Wassily want know why Piet read novel very quickly
'Wassily wonders why Piet reads novels quickly.'
b. ??**Kàn xiǎoshuō** Wassily xiǎng zhīdào [wèishéme Piet kàn de hěn kuài].
read novel Wassily want know why Piet read DE very quickly
Int.: 'As for reading novels, Wassily wonders why Piet reads them quickly.'

(18) *Complex NP-islands for high verb doubling*

- a. Wassily jùjué [Piet **kàn** xiǎoshuō hěn kuài de shuōfǎ].
Wassily reject Piet read novel very quickly DE_{nom} claim.
'Wassily rejects the claim that Piet reads novels quickly.'
b. ***Kàn xiǎoshuō** Wassily jùjué [Piet kàn de hěn kuài de shuōfǎ].
read novel Wassily reject Piet read DE very quickly DE_{nom} claim.
Int.: 'As for reading novels, Wassily rejects the claim that Piet reads them quickly.'

As exemplified in (19a) and (19b), low verb doubling out of islands is also impossible. However, the positional restrictions on long-distance verb doubling we will argue for in Sect. 3.1 mean that both ungrammatical examples in (19) can be ruled out independently of moving across islands (see a similar line of reasoning in Lai 2021, 2024).

(19) *Islands for low verb doubling*

- a. *Wassily **kàn xiǎoshuō** xiǎng zhīdào [wèishéme Piet kàn de hěn kuài].
Wassily read novel want know why Piet read DE very quickly
Int.: ‘Wassily wonders why Piet reads novels quickly.’
- b. *Wassily **kàn xiǎoshuō** jùjué [Piet kàn de hěn kuài de shuōfǎ].
Wassily read novel reject Piet read DE very quickly DE_{nom} claim.
Int.: ‘Wassily rejects the claim that Piet reads novels quickly.’

Moving on to reconstruction effects, the examples of high verb doubling in (20-22) all support the VP-movement analysis. In (20a), the VP *huà tāzìjǐ-de-xiàoxiàng* ‘draw the portrait of himself_i’ has to reconstruct to the lower thematic domain so that the reflexive can be bound by *méi-rén_i* ‘nobody_i’. Similarly in (21a), obligatory VP-reconstruction means that an instance of *Piet_i* is c-commanded by the pronoun *tā_i*, thereby violating Principle C. The example in (22a) shows that ungrammaticality only results if the pronoun c-commands the lower VP-copy. Further embedding of the pronoun removes the Principle C violation.

(20) *Principle A reconstruction effects in verb doubling*

- a. **Huà tāzìjǐ-de-xiàoxiàng** méi-rén_i huà de hěn kuài.
draw oneself-POSS-portrait NEG-people draw DE very quickly
‘No one_i draws the portrait of himself_i quickly.’
- b. Méi-rén_i **huà tāzìjǐ-de-xiàoxiàng** huà de hěn kuài.
NEG-people draw oneself-POSS-portrait draw DE very quickly
‘No one_i draws the portrait of himself_i quickly.’

(21) *Principle C reconstruction effects in verb doubling*

- a. ***Huà Piet_i-de-xiàoxiàng** tā_i huà de hěn kuài.
draw Piet-POSS-portrait 3SG draw DE very quickly
*‘He_i draws the portrait of Piet_i quickly.’
- b. *Tā_i **huà Piet_i-de-xiàoxiàng** huà de hěn kuài.
3SG draw Piet-POSS-portrait draw DE very quickly
*‘He_i draws the portrait of Piet_i quickly.’

(22) *No Principle C violation in verb doubling*

- a. **Huà Piet_i-de-xiàoxiàng** tā_i-de-māmā huà de hěn kuài.
draw Piet-POSS-portrait 3SG-POSS-mother draw DE very quickly
‘His_i mother draws the portrait of Piet_i quickly.’
- b. Tā_i-de-māmā **huà Piet_i-de-xiàoxiàng** huà de hěn kuài.
3SG-POSS-mother draw Piet-POSS-portrait draw DE very quickly
‘His_i mother draws the portrait of Piet_i quickly.’

All the low verb doubling constructions in (20b), (21b) and (22b) show the same results as their counterparts. Nonetheless, since the doubled VP does not move across its binding antecedent, the reconstruction diagnostics are not directly supportive to suggest a phrasal movement approach to low verb doubling.⁸

The hypothesis we pursue is that both low and high verb doubling are constructed by the same mechanism of moving the VP to some higher positions. Though the conventional movement diagnostics given above do not provide direct arguments for the low case, the overall distribution of verb doubling shows that both constructions are subject to identical well-formedness conditions. Lai (2021, 2024) demonstrates that, without the postverbal adverbial element, neither high nor low verb doubling can be constructed, as illustrated in (23). This parallel suggests that both kinds of verb doubling do not differ from each other with respect to their syntactic derivations. Considering that there exists compelling evidence in favour of the movement approach to high verb doubling, we conclude that low verb doubling

⁸ The fact that low verb doubling does not receive enough direct support from movement diagnostics (compared to its high counterpart) leads to a few non-movement analyses, e.g., Huang (1982, 1992); Tang (1990); Paul (2002a); Fang and Sells (2007); Hsu (2008, 2013).

displays no fundamental difference from its high counterpart in this respect.

- (23) a. Kàn nà-běn xiǎoshuō Piet kàn *(de hěn kuài).
 read that-CL novel Piet read DE very quickly
 Int.: ‘As for reading the novel, Piet reads it.’ *High verb doubling (cf. (10b))
- b. Piet kàn nà-běn xiǎoshuō kàn *(de hěn kuài).
 Piet read that-CL novel read DE very quickly
 Int.: ‘Piet reads the novel quickly.’ *Low verb doubling (cf. (10c))

The remaining problem is to account for the spell-out of the lower VP-copy. We adopt the proposal from Meadows and Yan (2023, to appear) that, the realisation of the lower V is related to the morphophonological requirement of particles like *de*. In the literature there exists an idea that movement-chain resolution can be prevented by affixation (and other morphological operations as well), if affixes happen to adhere to the copy of movement-chain (e.g., Nunes 2004; Landau 2006; also see Cheng 2007). The proposal we adopt shares a similar line of thinking behind it, and we refer readers to Meadows and Yan (2023, to appear) for more details.

To sum up, we have shown that there are two kinds of verb doubling constructions in Mandarin Chinese, differing in the position of the fronted VP relative to the surface subject. We have argued that both constructions are derived by the same mechanism of VP-movement, and posited that they are triggered by features distributed separately in the clausal spine. In the next section, we will delve into verb doubling in cross-clausal contexts.

3 Improper verb doubling and clause size

In this section we show that different groups of embedding predicates give rise to different patterns of long-distance VP-fronting. We propose that the varying locality profiles of VP-fronting are due to size of their clausal complements in question, giving language-internal arguments for three sizes of complement clauses.

3.1 Long-distance verb doubling

Building on Huang (2022)’s classification of clausal complementation types in Mandarin Chinese, we identify three kinds of embedding contexts, demonstrated by three groups of predicates (24). The overall paradigm that we will go through below is generalised in (25).

- (24) *Three groups of embedding predicates in Mandarin Chinese*
 Group 1: *fāxiàn* ‘find out’, *fǒurèn* ‘deny’, *juéde* ‘feel’, *rènwéi* ‘think’, *zhīdào* ‘know’...
 Group 2: *bī(pò)* ‘force’, *qiǎngpò* ‘force’, *quàn* ‘urge, persuade’, *yǐnyòu* ‘lure’, *zhǐdǎo* ‘instruct’...
 Group 3: *chángshì* ‘try’, *dǎsuan* ‘intend’, *jìhuà* ‘plan’, *juéding* ‘decide’, *kāishǐ* ‘begin’...

- (25) *Full paradigm of verb doubling in different embedding contexts*

Embedding predicate group	Matrix high VD	Matrix low VD	Embedded VD
Group 1: ‘think’ + [...]	✓	✗	✓
Group 2: ‘force’ + [...]	✓	✗	✓
Group 3: ‘try’ + [...]	✓	✓	✓

We start the discussion with Group 1 embedding predicates, represented by *rènwéi* ‘think’ in (26). The baseline configuration (26a) reveals that *rènwéi* can take a complement clause containing verb doubling.⁹ However, once the VP undergoes long-distance movement, there arises a position-contrast between (26b) and (26c). Though VP-fronting to the matrix-initial position is still possible (26c), the same phrasal movement to the matrix-medial position (26b) is prohibited. Such a configuration exemplifies what we term *improper verb doubling*.

⁹ Note that what is contained here is an instance of low VD. It is also possible to construct high VD in this Group 1 embedded clause, namely, the VP *kàn nà-běn xiǎoshuō* ‘read the novel’ moves to the embedded left periphery preceding the embedded subject *Piet* (and following the matrix predicate *rènwéi*). The same configuration cannot be easily constructed in Group 2 and 3 embedding contexts, since there is no overt subject in those embedded clauses. As will become clearer soon, we attribute such a difference to the size of different complement clauses.

- (26) a. Wassily rènwéi [Piet **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily think Piet read that-CL novel read DE very quickly
 ‘Wassily thinks that Piet reads the novel quickly.’ ‘think’ + embedded VD ✓
- b. *Wassily **kàn nà-běn xiǎoshuō** rènwéi [Piet kàn de hěn kuài].
 Wassily read that-CL novel think Piet read DE very quickly
 Int.: ‘Wassily thinks that Piet reads the novel quickly.’ ‘think’ + matrix low VD ✗
- c. **Kàn nà-běn xiǎoshuō** Wassily rènwéi [Piet kàn de hěn kuài].
 read that-CL novel Wassily think Piet read DE very quickly
 ‘As for reading the novel, Wassily thinks that Piet reads it quickly.’ ‘think’ + matrix high VD ✓

The distinction between proper and improper verb doubling is also identified in the Group 2 embedding contexts, as illustrated by *qiǎngpò* ‘force’ in (27). Unlike their Group 1 counterparts seen above, Group 2 predicates take a surface object and a clausal complement simultaneously. As with (26a), the baseline configuration (27a) shows that verb doubling can be constructed within the complement clause taken by a Group 2 predicate. The same contrast regarding long-distance VP-fronting appears again between (27b) and (27c). That is, the cross-clausal VP-fronting is only able to target the clause-initial position (27c) rather than the clause-medial one (27b).¹⁰

- (27) a. Wassily qiǎngpò Piet [**kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily force Piet read that-CL novel read DE very quickly
 ‘Wassily forces Piet to read the novel quickly.’ ‘force’ + embedded VD ✓
- b. *Wassily **kàn nà-běn xiǎoshuō** qiǎngpò Piet [kàn de hěn kuài].
 Wassily read that-CL novel force Piet read DE very quickly
 Int.: ‘Wassily forces Piet to read the novel quickly.’ ‘force’ + matrix low VD ✗
- c. **Kàn nà-běn xiǎoshuō** Wassily qiǎngpò Piet [kàn de hěn kuài].
 read that-CL novel Wassily force Piet read DE very quickly
 ‘As for reading the novel, Wassily forces Piet to read it quickly.’ ‘force’ + matrix low VD ✓

We see a different locality profile of long-distance VP-fronting when we look at Group 3 embedding predicates, represented by *chángshì* ‘try’ in (28). The baseline configuration (28a) shows that, similar to its Group 1 and 2 counterparts, it is possible to construct verb doubling within the embedded clause taken by a Group 3 predicate.¹¹ (28b) and (28c) further demonstrate that long-distance VP-fronting can land either after or before the matrix subject (i.e., *Wassily*) in a Group 3 embedding context (cf. (26b) and (27b)), presenting a scenario in which the cross-clausal verb doubling is no longer confined to the matrix left periphery (cf. (26c) and (27c)).

- (28) a. Wassily chángshì [**kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily try read that-CL novel read DE very quickly
 ‘Wassily tries to read the novel quickly.’ ‘try’ + embedded VD ✓
- b. Wassily **kàn nà-běn xiǎoshuō** chángshì [kàn de hěn kuài].
 Wassily read that-CL novel try read DE very quickly
 ‘Wassily tries to read the novel quickly.’ ‘try’ + matrix high VD ✓
- c. **Kàn nà-běn xiǎoshuō** Wassily chángshì [kàn de hěn kuài].
 read that-CL novel Wassily try read DE very quickly
 ‘As for reading the novel, Wassily tries to read it quickly.’ ‘try’ + matrix low VD ✓

¹⁰ Since the Group 2 representative we choose is commonly regarded as an object control verb, one may wonder whether the same patterns of long-distance verb doubling still hold in the context of ECM verbs, such as *xiāngxìn* ‘believe’ or *qídài* ‘expect’. According to the data we collect, the result shows no difference from their object control counterparts. A potential concern here is whether these alleged ‘ECM’ verbs in Mandarin Chinese really behave identically to those in English, since there is recent literature (e.g., Ussery et al. 2016; Paul 2021) arguing against the existence of Mandarin ECM verbs. As this is a topic beyond the scope of this paper, we confine Group 2 predicates to object control verbs only throughout the following discussion.

¹¹ Some native speakers report to us that the meaning of (28a) sounds pragmatically awkward to them. However, presented with the scenario below, the acceptance of (28a) improves significantly: Wassily used to read the novel in a very slow manner. Now he wants to change his way of reading, trying to read the novel as quickly as possible, or at least more quickly than before.

To summarise, we have shown that long-distance VP-fronting differs in targeting the two landing sites in various embedding contexts. Though it is always possible for the VP to move across the complement clause and target the matrix-initial position, it becomes more restricted when the VP targets the relatively lower, medial position in the matrix clausal spine, giving rise to the proper/improper verb doubling contrast. Viewed in this way, the embedding predicate groups that we have identified in the first place can also be divided only into two: one group (including Group 1 and 2) exhibits improper verb doubling, whereas another group (i.e., Group 3 only) does not show the same pattern. So it is worth questioning why and how we recognise three instead of two predicate groups. As will be clearer in Sect. 4, we argue that whether improper verb doubling is attested reflects the size of clausal complement. The fact that we have seen in this section indicates that both Group 1 and 2 predicates take bigger complements than their Group 3 counterparts. It is the empirical findings we use to diagnose clause size in Sect. 3.2 that lead us to the distinction of three predicate groups as presented here.

3.2 Three sizes of clausal complement

To capture the locality differences presented above, we rely on the idea that clausal complements vary in size. In the literature of Mandarin syntax, clause-size variation has been proposed and discussed in, for instance, Xue and McFetridge (1996, 1998), Grano (2012, 2015, 2017), and Huang (2022), among many others. Along the same lines, we propose that Group 1 predicates take CP-sized complements, Group 2 TP-sized ones, and Group 3 FP-sized ones. The proposed size difference of clausal complements taken by different groups of embedding predicates is schematised in (29). In what follows we offer two suggestive empirical arguments in this direction.

(29) *Embedding predicates and their corresponding clausal complements*

- a. Pred._{Group 1} + [CP [MP^{epis} [TP [MP^{deon} [FP [VoiceP ...]]]]]]
- b. Pred._{Group 2} + [TP [MP^{deon} [FP [VoiceP ...]]]]
- c. Pred._{Group 3} + [FP [VoiceP ...]]

3.2.1 Object shift in embedding contexts

The first argument comes from *object shift* (OS), a well-known displacement phenomenon that is widely discussed in the Mandarin literature (e.g., Ernst and Wang 1995; Shyu 1995; Paul 2002b, 2005; Chen 2023, to appear). We show that Group 1 clausal complements can provide more positions for object shift than their Group 2 and 3 counterparts, which we argue reflects reduced clause structure. The key patterns are summarised in (30), regarding the distribution of object shift relative to TP within the embedded clause.

(30) *Object shift in different embedding contexts*

Embedding predicate group	OS above TP _{Emb}	OS below TP _{Emb}	Clause size
Group 1: ‘think’ + [...]	✓	✓	CP-sized
Group 2: ‘force’ + [...]	?	✓	TP-sized
Group 3: ‘try’ + [...]	✗	✗	FP-sized

Our discussion begins with the complement clauses taken by Group 1 predicates. As shown in (31), the object *nà-běn xiǎoshuō* ‘the novel’, which is canonically understood to be selected by the verb *kàn* ‘read’, can be displaced to the left of the verb and show up either before the embedded subject *Piet* (31a) or after it (31b). The displaced object receives a topicalised interpretation in both positions (indicated by the words in capitals).

- (31) a. Wassily rènwéi [Piet **nà-běn xiǎoshuō** *kàn de hěn kuài*].
Wassily think Piet that-CL novel read DE very quickly
‘Wassily thinks that Piet reads THAT NOVEL quickly.’
- b. Wassily rènwéi [**nà-běn xiǎoshuō** Piet *kàn de hěn kuài*].
Wassily think that-CL novel Piet read DE very quickly
‘Wassily thinks that, THAT NOVEL, Piet reads quickly.’

Chen (2023, to appear), following Paul (2002b, 2005), proposes that in Mandarin Chinese, projections in the domain below TP (= IP in her usage) parallel those in the articulated left periphery (Rizzi 1997). According to Chen, the

shifted object is actually the consequences of DP movement that targets either TP-internal [Spec, TopP] (as in (31a)) or TP-external [Spec, TopP] (as in (31b)). As schematised in (32), that both kinds of topicalisation are available in Group 1 complement clauses follows our proposal that Group 1 predicates take full-fledged, CP-sized clausal complements.

- (32) *Object shift in the Group 1 embedding context*
 Pred._{Group 1} + [CP <OS> [TP Subj <OS> ...]]

Object shift is also attested in complement clauses taken by Group 2 predicates, as exemplified in (33). As with its counterpart in the Group 1 embedding context, the shifted object receives a topicalised interpretation, suggesting that the displacement of the object is triggered by the same mechanism of topicalisation in such a case as well.

- (33) Wassily qiǎngpò Piet [nà-běn xiǎoshuō kàn de hěn kuài].
 Wassily force Piet that-CL novel read DE very quickly
 ‘Wassily forces Piet to read THAT NOVEL quickly.’

The difference of the Group 2 embedding context is that there is no overt embedded subject in the construction, leading to the puzzle whether the shifted object lands within the TP- or the CP-domain.¹² Putting aside this problem for a while (which we will return to very soon), the fact that Group 2 complements can host object shift indicates that they are at least TPs, as schematised in (34).

- (34) *Object shift in the Group 2 embedding context*
 Pred._{Group 2} + [TP OS ...]

Unlike what we see in the previous two situations, object shift is not acceptable in Group 3 complement clauses, as shown in (35).

- (35) *Wassily chángshì [nà-běn xiǎoshuō kàn de hěn kuài].
 Wassily try that-CL novel read DE very quickly
 Int.: ‘Wassily tries to read THAT NOVEL quickly.’

The ungrammaticality of (35) implies that complement clauses taken by Group 3 predicates are smaller than their Group 1 and 2 counterparts. Comparing with the case of Group 2 complements seen in (33), we conclude that Group 3 clausal complements are not as big as TP. Recall that embedded verb doubling is possible within the Group 3 context (e.g., (28a)), and that low verb doubling is proposed to land in [Spec, FP]. We postulate that Group 3 complement clauses are FP-sized, and that FP is lower than TP in the clausal spine. That no object shift is attested in Group 3 complements is illustrated in (36).

- (36) *No object shift in the Group 3 embedding context*
 Pred._{Group 3} + [FP (*OS) ...]

Variation in clause size seems to provide a direct account for the distribution of object shift within different embedding contexts. Further evidence for our account comes from the distribution of adverbs and their interaction with the shifted object. Let us first have a look at another pair of object shift cases within the Group 1 embedding context that co-occur with different adverbs respectively, as given in (37).

- (37) a. Wassily rènwéi [CP [TP Piet nà-běn xiǎoshuō yòu kàn-le sān biàn]].
 Wassily think Piet that-CL novel again read-PRF three time

¹² Object shift cannot target the position between Group 2 predicate and its surface object, as given in ((i)).

- (i) *Wassily qiǎngpò nà-běn xiǎoshuō Piet [kàn de hěn kuài].
 Wassily force that-CL novel Piet read DE very quickly
 Int: ‘Wassily forces Piet to read THAT NOVEL quickly.’

The ungrammatical result is actually what we predict, since Group 2 predicates are all object control verbs, and there is no available position to host object shift before the object.

‘Wassily thinks that Piet again read THAT NOVEL three times.’

- b. Wassily rènwéi [_{CP} **nà-běn xiǎoshuō**, *xìngkuī* [_{TP} Piet kàn-le sān biàn]].
 Wassily think that-CL novel fortunately Piet read-PRF three time
 ‘Wassily thinks that, THAT NOVEL, fortunately Piet read three times.’

(37a) presents a configuration where the TP-internal object shift is higher than *yòu* ‘again’, a VP-level adverb in the sense of Paul (2017) (cf. Ernst 2014). According to Paul, such adverbs refer to as well as modify the internal structure of an event: in (37a) *yòu* is applied to the event of ‘reading that novel three times’, thus the whole event gets repeated and happens again. The positions of these so-called ‘VP-level’ adverbs are generally assumed to be relatively low in the *fseq* (Cinque 1999). Here we simply presume that *yòu* marks the edge of the event/verbal domain which is for sure lower than TP.¹³

(37b) exemplifies an instance of TP-external object shift that precedes not only the embedded subject *Piet* but also a sentence-level adverb *xìngkuī* ‘fortunately’. The label ‘sentence-level’, as defined by Paul (2017), actually consists of a number of sub-categories (e.g., as proposed in Ernst 2014). Though most sentence-level adverbs are liable to occur both to the right and left of the subject, a few of them are confined to the clause-initial position. Paul (2017) claims that *kěxī* ‘unfortunately’ is the only candidate for the latter group, which is confined to clause-initial and excluded from clause-medial position. However, we find that the latter position for *kěxī* is marginally acceptable. *Xìngkuī* ‘fortunately’ shares the same distribution with *kěxī* according to our data: it is definitely preferred in the clause-initial position. This observation is crucial because it would be very likely for any constituent to the left of *xìngkuī* (as well as *kěxī*) to be automatically assumed to occur in the clausal left periphery, just like the shifted object in (37b).¹⁴ Based on (37), we schematise the distribution of object shift relative to distinct types of adverbs in the Group 1 embedding context as follows:

- (38) *Object shift in the Group 1 embedding context (relative to adverbs)*
 Pred._{Group 1} + [_{CP} <OS> Adv_{S-Level} [_{TP} Subj <OS> Adv_{VP-Level} ...]]

Proceeding to the Group 2 embedding context, we first see that the sentence-level adverb *xìngkuī* is not allowed to appear independently in the Group 2 embedding context, as shown in (39).

- (39) *Wassily qiǎngpò Piet [_{TP} *xìngkuī* kàn-le **nà-běn xiǎoshuō** sān biàn].
 Wassily force Piet fortunately read-PRF that-CL novel three time

The ungrammatical result follows naturally our proposal that Group 2 clausal complements are generally TP-sized. Since the left periphery is totally absent within the Group 2 embedding context, there exists no position in the clausal spine for sentence-level adverbs like *xìngkuī* to be hosted properly. Furthermore, in this case, no object shift is predicted to target a position that is even higher than the sentence-level adverbs. This is borne out by (40).

- (40) *Wassily qiǎngpò Piet [_{TP} **nà-běn xiǎoshuō** *xìngkuī* kàn-le sān biàn].
 Wassily force Piet that-CL novel fortunately read-PRF three time

By comparison, the shift object is still able to co-occur with the VP-level adverb *yòu*, as given in (41). This is because object shift can target the position that is within the TP but higher than the entire event/verbal domain including the position of *yòu*.

- (41) Wassily qiǎngpò Piet [_{TP} **nà-běn xiǎoshuō** *yòu* kàn-le sān biàn].
 Wassily force Piet that-CL novel again read-PRF three time
 ‘Wassily forced Piet to again read THAT NOVEL three times.’

¹³ As for the exact position of *yòu*, Liu (2021, 2024) proposes that it can be base-generated with an unvalued Asp feature in [Spec,vP], which would, for the sake of specifier-head agreement, further move to the specifier position of the closet AspP, a position he takes to be lower than TP but higher than vP. Liu’s proposal seems compatible with our assumption. However, since his analysis lays the foundation on the ambiguous interpretations of *yòu*, which do not concern us here, we maintain agnostic towards its precise position.

¹⁴ Note that the shifted object can also occur to the right of *xìngkuī* in (37b), which follows Rizzi (1997)’s proposal that there are two topic positions in the left periphery. By contrast, it is impossible for object shift to appear at the right side of *yòu* in (37a), which is compatible with our assumption that *yòu* stays at the edge of verbal domain lower than TP. The same fact also implies that the position of *yòu* is too low to be included in the TP-domain.

It is worth pointing out that our account does not empirically rule out the possibility that the shifted object in (41) in effect lands in a TP-external position that is lower than *xìngkuī* (see fn. 14), which is why we give a question marker to the corresponding space in Table (30). If that was the actual situation, one needs to come up with some different assumption in order to explain why the independent appearance of *xìngkuī* is impossible in (39). This possibility can be ruled out by assuming that the TP-external shifted object may act as an intervenor between the surface object and its controlled PRO in the embedded clause. At the moment we simply conclude that Group 2 complement clauses are TP-sized as represented in (42).

- (42) *Object shift in the Group 2 embedding context (relative to adverbs)*
 Pred._{Group 2} + [TP **OS** Adv_{VP-Level} ...]

Moving on to Group 3 complement clauses, it is not a surprise that object shift is unable to co-occur with either the VP-level adverb *zài* ‘again’ (43a) or the sentence-level one *xìngkuī* (43b), since this specific kind of DP-movement is prohibited in the Group 3 embedding context at the outset (as seen in (35)), leading to the hypothesis that Group 3 complements are FP-sized only (as represented in (36)). Such an idea is reinforced by the fact that the sentence-level adverb is also unable to show up on its own in the Group 3 embedding context (44).

- (43) a. *Wassily chángshì [FP **nà-běn xiǎoshuō** *zài* kàn sān biàn].
 Wassily try that-CL novel again read three time
 ‘Wassily tries to again read THAT NOVEL three times.’
 b. *Wassily chángshì [FP **nà-běn xiǎoshuō** *xìngkuī* kàn sān biàn].
 Wassily try that-CL novel fortunately read three time
 (44) *Wassily chángshì [FP *xìngkuī* kàn **nà-běn xiǎoshuō** sān biàn].
 Wassily try fortunately read that-CL novel three time

Interestingly, as exemplified in (45), the VP-level adverb is able to co-occur with the embedded verb doubling, contrary to the configuration involving embedded object shift as in (43a).

- (45) Wassily chángshì [FP *zài* **kàn nà-běn xiǎoshuō** kàn sān biàn].
 Wassily try again read that-CL novel read DE very quickly
 ‘Wassily tries to again read THAT NOVEL three times.’

Empirically, the contrast between (45) and (43a) indicates clearly that low verb doubling and TP-internal object shift target different positions in the clausal spine: low verb doubling can appear below the VP-level adverb, whereas object shift (in all possible cases) has to occur above the same adverb. Theory-wise, such a contrast further confirms the idea that FP is lower than TP in the clausal spine (recall Sect. 2.2): [Spec, FP] is targeted by low verb doubling rather than TP-internal object shift, as schematised in (46). We will rely on this important idea while extending our Level Embedding approach to long-distance object shift in Sect. 5.

- (46) *Verb doubling but no object shift in the Group 3 embedding context (relative to adverbs)*
 Pred._{Group 3} + [FP (***OS**) Adv_{VP-Level} **VD** ...]

To recap our argument, we have shown that the distribution of embedded object shift varies in different embedding contexts, and argued that it is the clause-size variation that determines the result of displacing the object. The key patterns of object shift as well as the proposed clause sizes are summarised again in (47), repeated from (30).

- (47) *Object shift in different embedding contexts*
- | Embedding predicate group | OS above TP _{Emb} | OS below TP _{Emb} | Clause size |
|----------------------------|----------------------------|----------------------------|-------------|
| Group 1: ‘think’ + [...] | ✓ | ✓ | CP-sized |
| Group 2: ‘force’ + [...] | ? | ✓ | TP-sized |
| Group 3: ‘try’ + [...] | ✗ | ✗ | FP-sized |

3.2.2 Modal hierarchy in embedding contexts

The second argument for variation in clause size focuses on the distribution of various modals in complement clauses. In what follows we show that Group 1 clausal complements display a full range of modals within the embedding context, whereas their Group 2 and 3 counterparts are only able to exhibit a reduced range of, or even a complete absence of, the same modals. We argue such a contrast again reflects the reduced clause structure of Group 2 and 3 complements.

One characteristic of modals in Mandarin Chinese is that they display rigid word orders when they co-occur (Lin 2011, 2012; Lai and Li 2023, a.o.).¹⁵ For instance, as shown in (48a), the epistemic modal *yīnggāi* ‘should’ must precede another epistemic modal *kěnéng* ‘may’ when they appear together. When both epistemic modals respectively co-occur with the future modal *huì*, either of them precedes the future one, as given in (48b) and (48c). If these three appear together in the same configuration, such as (48d), their relative positions to each other remain the same as what we have seen above, that is, the epistemic ‘may’ is preceded by the epistemic ‘should’ and followed by the future ‘will’.

- (48) a. Piet *yīnggāi*^{epis} *kěnéng*^{epis} líkāi lúndūn le.
 Piet should may leave London SFP
 ‘It should be the case that Piet may have left London.’ should^{epis} >> may^{epis}
- b. Piet *yīnggāi*^{epis} *huì*^{future} líkāi lúndūn.
 Piet should will leave London
 ‘It should be the case that Piet will leave London.’ should^{epis} >> will^{future}
- c. Piet *kěnéng*^{epis} *huì*^{future} líkāi lúndūn.
 Piet may will leave London
 ‘It may be the case that Piet will leave London.’ may^{epis} >> will^{future}
- d. Piet *yīnggāi*^{epis} *kěnéng*^{epis} *huì*^{future} líkāi lúndūn.
 Piet should may will leave london
 ‘It should be the case that it is likely that Piet will leave London.’ should^{epis} >> may^{epis} >> will^{future}

Another rigid sequence of word order can be found in cases involving deontic modals relative to epistemic ones. As given in (49a), the deontic modal *bìxū* ‘must’ can appear before another deontic modal *děi* ‘have to’.¹⁶ When either of them co-occurs with an epistemic modal, e.g., *kěnéng* ‘may’, the latter has to be positioned before the former, as seen respectively in (49b) and (49c). When these two deontic modals show up with the epistemic one at the same time, they are both preceded by the epistemic modal, with the internal order between them unchanged, as illustrated in (49d).

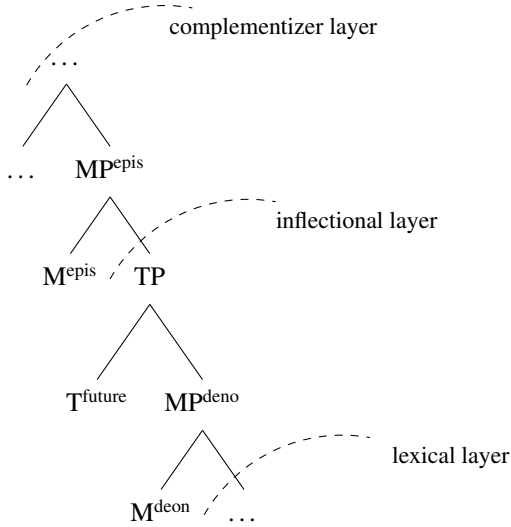
- (49) a. Piet *bìxū*^{deon} *děi*^{deon} líkāi lúndūn.
 Piet must have-to leave London
 ‘Piet must have to leave London.’ must^{deon} >> have to^{deon}
- b. %Piet *kěnéng*^{epis} *bìxū*^{deon} líkāi lúndūn le.
 Piet may must leave London SFP
 ‘It may be the case that Piet must leave London.’ may^{deon} >> must^{deon}
- c. Piet *kěnéng*^{epis} *děi*^{deon} líkāi lúndūn le.
 Piet may have-to leave London SFP
 ‘It may be the case that Piet has to leave London.’ may^{epis} >> have to^{deon}
- d. Piet *kěnéng*^{epis} *bìxū*^{deon} *děi*^{deon} líkāi lúndūn le.
 Piet may must have-to leave London SFP
 ‘It may be the case that Piet must have to leave London.’ may^{epis} >> must^{deon} >> have to^{deon}

¹⁵ Throughout the discussion, we remain agnostic towards the actual category of modals, since in Mandarin it is not immediately obvious whether a modal is verbal or adverbial. What matters to us is that every modal is licensed by a unique functional projection, and the presence/absence of each modal is contingent on the existence of its corresponding functional projection in the clausal *fseq*.

¹⁶ There are two points worth mentioning here. First, the meaning difference between *bìxū* and *děi* is actually quite trivial. The different translations that we give to these two deontic modals only serve the purpose of tearing them apart. Second, though we agree that *bìxū* preceding *děi* is the preferred order in general, we find the reverse word order acceptable as well. However, the flexible word order between deontic modals, if it holds, would not undermine the point we want to raise here, which is, epistemic modals always precede deontic modals.

Tsai (2015) claims that the rigid word order among modals in Mandarin, as seen above, can be aligned with the syntactic configuration advocated in the cartographic fashion (Rizzi 1997; Cinque 1999). Assuming that the future modal *huì* acts as the head of TP, he proposes that epistemic modals, which are structurally higher than the future, are situated in the complementizer layer outside the TP. As for deontic modals, Tsai posits that they are lower than the future modal, but still located in the same inflectional layer with the future. The modal hierarchy in the sense of Tsai (2015) is simply illustrated in (50) (cf. Lin 2012):

(50) *Modal hierarchy in Mandarin Chinese (in the sense of Tsai (2015))*



With this hierarchy in mind, we start looking at the distribution of modals in different embedding contexts, with a focus on epistemic and deontic modals. As illustrated in (51), the modals that we concern are all compatible with embedded verb doubling in Group 1 complement clauses. (51a) and (51b) respectively show the configurations involving the epistemic *yīnggāi* and *kěnéng*. (51c) and (51d) are cases where the deontic *bìxū* and *děi* occur in the embedded clauses. Note that, to better accommodate to the meaning of deontics, the postverbal adverbial in the latter two examples is slightly different from its counterpart in the former two, conveying a comparative state that has yet to achieve (i.e., *gèng kuài yīxiē* ‘more quickly’). The empirical picture presented in (51) is in line with our proposal regarding the clause size: since Group 1 predicates are proposed to take CP-sized complement clauses, it follows that all the modals in principle are able to occur in such configurations.¹⁷

- (51) a. Wassily *rènwéi* [_{CP} Piet *yīnggāi*^{epis} **kàn nà-běn xiǎoshuō** *kàn de hěn kuài*].
Wassily think Piet should read that-CL novel read DE very quickly
‘Wassily thinks it should be the case that Piet reads the novel quickly.’ should^{epis} ✓
- b. Wassily *rènwéi* [_{CP} Piet *kěnéng*^{epis} **kàn nà-běn xiǎoshuō** *kàn de hěn kuài*].
Wassily think Piet may read that-CL novel read DE very quickly
‘Wassily thinks it may be the case that Piet reads the novel quickly.’ may^{epis} ✓
- c. Wassily *rènwéi* [_{CP} Piet *bìxū*^{deon} **kàn nà-běn xiǎoshuō** *kàn de gèng kuài yīxiē*].
Wassily think Piet must read that-CL novel read DE more quickly a-bit
‘Wassily thinks that Piet must read the novel more quickly.’ must^{deon} ✓
- d. Wassily *rènwéi* [_{CP} Piet *děi*^{deon} **kàn nà-běn xiǎoshuō** *kàn de gèng kuài yīxiē*].
Wassily think Piet will read that-CL novel read DE more quickly a-bit
‘Wassily thinks that Piet has to read the novel more quickly.’ have to^{deon} ✓

¹⁷ In the relevant literature, both Lin (2011, 2012) and Zhang (2019) argue that epistemic modals (and future modals) may occur in finite complements. Such an idea is theoretically compatible with our proposal, as CP-sized complements are generally considered finite. However, we do not have too much to comment on the relation between finiteness and clause size in this paper. We refer readers to the references cited above as well as recent work by Huang (2022).

The difference with respect to the modal hierarchy starts appearing in Group 2 clausal complements. As shown in (52a) and (52b) separately, neither the epistemic *yīnggāi* nor the epistemic *kěnéng* is allowed to exist in the Group 2 embedding context, contrary to their status in the Group 1 context (cf. (51a) and (51b)). The deontic *bìxū* and *děi*, however, are still able to show up in Group 2 complement clauses, as given in (52c) and (52d). Under our proposal that Group 2 complements are TP-sized, such a contrastive result is in fact what we would expect. Recall that in the modal hierarchy (50), epistemic modals are higher than TP and assumed to be excluded from the inflectional layer, whereas deontic modals are argued to be lower than TP and inside the inflectional layer. Thus, the contrast between epistemics and deontics within the TP-sized, Group 2 embedding context fits neatly with the division of layers in the modal hierarchy.

- (52) a. *Wassily qiǎngpò Piet [_{TP} *yīnggāi*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily force Piet should read that-CL novel read DE very quickly
should^{epis} ✗
- b. *Wassily qiǎngpò Piet [_{TP} *kěnéng*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily force Piet may read that-CL novel read DE very quickly
may^{epis} ✗
- c. Wassily qiǎngpò Piet [_{TP} *bìxū*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
Wassily force Piet must read that-CL novel read DE more quickly a-bit
‘Wassily forced Piet to have to read the novel more quickly.’
must^{deon} ✓
- d. %Wassily qiǎngpò Piet [_{TP} *děi*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
Wassily force Piet must read that-CL novel read DE more quickly a-bit
‘Wassily forced Piet to have to read the novel more quickly.’
have to^{deon} ✓

The modal hierarchy continues to ‘vanish’ within the Group 3 embedding context. As shown in (53), all the epistemic and deontic modals that we concern are unable to occur in any complement clause taken by Group 3 predicates. By comparison, if the modals are removed from the embedded clauses, all of them become grammatical, exhibiting embedded verb doubling as we have already seen in section 3.1 (e.g., (28a)). The results in (53) suggest that Group 3 complement clauses, which we propose are only FP-sized, might be structurally too small to host any modals that we discuss here.

- (53) a. *Wassily chángshì [_{FP} *yīnggāi*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily try may read that-CL novel read DE very quickly
should^{epis} ✗
- b. *Wassily chángshì [_{FP} *kěnéng*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily try may read that-CL novel read DE very quickly
may^{epis} ✗
- c. *Wassily chángshì [_{FP} *bìxū*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
Wassily try must read that-CL novel read DE more quickly a-bit
must^{deon} ✗
- d. *Wassily chángshì [_{FP} *děi*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
Wassily try must read that-CL novel read DE more quickly a-bit
have to^{deon} ✗

The patterns regarding the distribution of modals in clausal complements are summarised in (54). With the modal hierarchy (50) taken into account, we have argued that the distributional patterns can receive a direct explanation from our proposal. That is, whether a modal shows up in a complement clause depends on the size of the clause it enters.

- (54) *Modal hierarchy in different embedding contexts*

Embedding predicate group	Should ^{epis}	May ^{epis}	Must ^{deon}	Have to ^{deon}	Clause size
Group 1: ‘think’ + [...]	✓	✓	✓	✓	CP-sized
Group 2: ‘force’ + [...]	✗	✗	✓	✓	TP-sized
Group 3: ‘try’ + [...]	✗	✗	✗	✗	FP-sized

3.3 Note on types of clausal complementation

At the beginning of Sect. 3.1, we mentioned that the predicate groups we identify are built on the clause types proposed by Huang (2022). In the recent literature, He (2020, 2024) and Huang (2022), among many others, propose that the notion of finiteness can be viewed as part of the *Implicational Complementation Hierarchy* (ICH) proposed by Wurmbrand and Lohninger (2023). Along the ICH-based typology, Huang (2022:22-24) recognises three types of verbal complements in Mandarin Chinese, characterising Type I as ‘the most finite’, Type II as ‘somewhat infinitival’, and Type III as ‘the most infinitival’ (see a similar proposal in He 2020, 2024).

There are overlaps between Huang’s clause types and our predicate groups. To be precise, Huang’s Type I is our Group 1, and his Type III falls into our Group 3. The actual difference lies in his Type II versus our Group 2: only those Type II predicates which are able to take a surface object plus a clausal complement (i.e., object control verbs) are defined as our Group 2 predicates, and the remaining Type II ones all enter our Group 3. The correspondence between these two ways of classifying predicates/clauses is illustrated in (55).

(55) *Correspondence between our predicate groups and Huang’s clause types*

Predicate group proposed in here	Clause type proposed in Huang (2022)
Group 1	Type I
Group 2	Type II
Group 3	Type II & III

The question that arises immediately is why Group 3 corresponds to both Type II and III. We argue that this mismatch comes as a natural result of classifying predicates and clauses based on different criteria. Recall that our evidence to distinguish predicate groups as well as their corresponding embedding context is primarily those patterns that can be directly explained by varying clause size. The distinction of clause types, however, is originally motivated by the meaning of their embedding predicates (see Wurmbrand and Lohninger 2023 for details). The two different criteria together lead us to expect there should be predicates that syntactically behave identically to other Group 3 and semantically select clausal complements of Type II. In what follows we use *dǎsuàn* ‘plan’ as the representative to illustrate the property of this specific group of predicates.

As discussed by both He (2020, 2024) and Huang (2022), one defining characteristic of differentiating clause types is their temporal independence (see a summarised overview in He 2024:280). Type I complements can flexibly have independent time adverbs in each of the clauses, as shown in (56). Though Type II complements also permit time adverbs different from those in the matrix, the former must be constrained by the latter in terms of their temporal relation, as seen in (57) and (58). Type III complements are the most temporally dependent, which forbid the co-occurrence of different time adverbs at all, as given in (59). In this respect, it is reasonable to label *dǎsuàn* as Type II (as with *qiǎngpò*) rather than Type III (as with *chángshì*).

- (56) a. **Zuótiān** Wassily rènwéi [Piet **jīntiān** huì líkāi lúndūn].
yesterday Wassily think Piet today will leave London
‘Yesterday Wassily thought that Piet will leave London today.’
- b. **Zuótiān** Wassily rènwéi [Piet **shàng-xīngqí** líkāi-le lúndūn].
yesterday Wassily think Piet last-week leave-PFV London
‘Yesterday Wassily thought that Piet left London last week.’
- (57) a. **Zuótiān** Wassily qiǎngpò Piet [**jīntiān** líkāi lúndūn].
yesterday Wassily force Piet today leave London
‘Yesterday Wassily forced Piet to leave London today.’
- b. ***Zuótiān** Wassily qiǎngpò Piet [**shàng-xīngqí** líkāi lúndūn].
yesterday Wassily force Piet last-week leave London
‘*Yesterday Wassily forced Piet to leave London last week.’

- (58) a. **Zuótiān** Wassily dǎsuàn [**jīntiān** líkāi lúndūn].
yesterday Wassily plan today leave London
‘Yesterday Wassily planned to leave London today.’
- b. ***Zuótiān** Wassily dǎsuàn [**shàng-xīngqí** líkāi lúndūn].
yesterday Wassily plan last-week leave London
‘*Yesterday Wassily planned to leave London last week.’
- (59) a. ***Zuótiān** Wassily chángshì [**jīntiān** líkāi lúndūn].
yesterday Wassily try today leave London
‘*Yesterday Wassily tried to leave London today.’
- b. ***Zuótiān** Wassily chángshì [**shàng-xīngqí** líkāi lúndūn].
yesterday Wassily try last-week leave London
‘*Yesterday Wassily tried to leave London last week.’

The picture becomes different when checking all the patterns we have seen in this section on *dǎsuàn*. It turns out that *dǎsuàn* demonstrates all the same patterns as *chángshì* (a Group 3 predicate in our sense) instead of *qiǎngpò* (our Group 2 predicate): both kinds of long-distance verb doubling are attested (60) (cf. (28)); the judgement of embedded object shift is relatively degraded (61) (cf. (35)); and modal hierarchy is totally absent in its clausal complements (62) (cf. (53)).

- (60) a. Wassily dǎsuàn [**kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
Wassily plan read that-CL novel read DE more quickly a-bit
‘Wassily plans to read the novel more quickly.’ ‘plan’ + embedded VD ✓
- b. Wassily **kàn nà-běn xiǎoshuō** dǎsuàn [kàn de gèng kuài yīxiē].
Wassily read that-CL novel plan read DE more quickly a-bit
‘Wassily plans to read the novel more quickly.’ ‘plan’ + matrix high VD ✓
- c. **Kàn nà-běn xiǎoshuō** Wassily dǎsuàn [kàn de gèng kuài yīxiē].
read that-CL novel Wassily plan read DE more quickly a-bit
‘As for reading the novel, Wassily plans to read it more quickly.’ ‘plan’ + matrix low VD ✓
- (61) ??Wassily dǎsuàn [**nà-běn xiǎoshuō** kàn de gèng kuài yīxiē / sān biàn].
Wassily plan that-CL novel read DE more quickly a-bit three time
Int.: ‘Wassily plans to read THAT NOVEL more quickly / three times.’
- (62) a. *Wassily dǎsuàn [yīnggāi^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily plan may read that-CL novel read DE very quickly
should^{epis} ✗
- b. *Wassily dǎsuàn [kěnéng^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily plan may read that-CL novel read DE very quickly
may^{epis} ✗
- c. *Wassily dǎsuàn [bìxū^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
Wassily plan must read that-CL novel read DE more quickly a-bit
must^{deon} ✗
- d. *Wassily dǎsuàn [děi^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
Wassily plan must read that-CL novel read DE more quickly a-bit
have to^{deon} ✗

The comparisons made above clearly demonstrate that the predicate group in our sense and the clause type in Huang (2022)’s sense are not the same concept due to their different criteria of classification. Therefore it does not constitute a huge surprise that our predicate groups and Huang’s clause types do not exhibit one-on-one correspondence. Nevertheless, we still face the dilemma that Type II clauses seem to correspond to two different sizes, namely,

TP taken by Group 2 predicates and FP taken by Group 3. Here we simply suggest two potential accounts. The first one is that, since neither Huang’s classification nor ours is exhaustive, there might be more predicate groups/clause types in Mandarin Chinese than the tripartite analysis presented in either work. That is to say, verbs like *dǎsuàn* might constitute a group on its own independently of other Group 3 predicates because of the distinct semantic requirements on their complements. Another possible explanation is that semantic distinctions like temporal independence might not be a good diagnostic for tearing apart infinitives, since tense in infinitives is always constrained to some extent by the embedding predicates. We leave both possibilities for future research, and stick to our tripartite classification of predicates in the rest of the paper.

3.4 Improper verb doubling as a Height-Locality effect

This section has so far outlined what Keine (2016, 2019, 2020) calls a *Height-Locality effect* as generalised in (63): VP-movement to a lower position in the clause is possible out of smaller range of complement clauses than VP-movement to the clausal periphery. Mandarin Chinese is therefore part of an increasingly diverse range of languages in which such effects have been observed in movement dependencies.¹⁸

(63) *Height-Locality effect*

A syntactic dependency whose locality appears to be conditioned by the position of its trigger (or landing site) in the clausal functional sequence. [based on Keine 2020:11-15, *et seq.*]

In English this most obviously concerns restrictions on raising-to-subject, which cannot proceed out of finite clausal complements (e.g., (64a) vs. (64b)).¹⁹ This contrasts with the behaviour of, e.g., *wh*-movement, which can escape finite complements (64c). What this contrast seems to track is the difference in landing site: *wh*-movement and raising-to-subject target different linear and structural positions ([Spec, CP] vs. [Spec, TP]). In other words, *wh*-movement lands higher in the clausal spine than raising, and it is less bounded than raising.

(64) *Height-Locality effects in English raising-to-subject*

- a. Anouk seems to like banana bread.
- b. *Anouk seems likes banana bread.
- c. Who does it seem likes banana bread?

In the growing cross-linguistic picture of Height-Locality effects, various movement types beyond raising exhibit clause-boundedness without the full range of archetypal A-movement properties, as given in (65). To our knowledge, Height-Locality effects with movement have up to now only been observed with the movement of nominals. The constraints on VP-movement in Mandarin Chinese show us that VP-movement is not exceptional in this regard. We go on to show that Height-Locality effects with nominal movement also exist in Mandarin, with reference to long-distance object shift in Sect. 5.

(65) *Height-Locality effects beyond raising*

Scrambling in Hindi	(Keine 2016, 2020)
<i>Wh</i> -movement out of V-final vs. V2 complements in German	(Keine 2016, 2020)
Particle movement in Hungarian	(Egressy 2023)
<i>Wh</i> -movement in Georgian	(Bondarenko 2024)
Movement forming relative clauses in Swahili	(Meadows 2024)

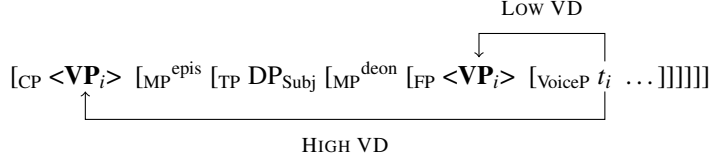
With this cross-linguistic context clear, let us round out this section. We had two goals: (i) present the basic descriptive facts about long-distance verb doubling, and (ii) motivate an analysis of these facts in which a key variable is the size of the embedded clausal complement. We presented straightforwardly syntactic arguments, from nominal movement within embedded clauses and the distribution of modals, that complements come in three sizes. What

¹⁸ We set aside the issue of how this manifests with cases discussed extensively by sub-extraction and remnant movement (e.g., Abels 2007, 2009; Neeleman and van de Koot 2010; Egressy 2023), as well domains beyond movement where these effects may also hold, such as agreement (Keine 2016, 2019, 2020) and case-assignment (Poole 2022).

¹⁹ Depending on the exact analysis of the constructions in question, it also bears on the clause-bound dependencies in infinitival relatives and *tough*-constructions (see, e.g., Douglas 2016; Longenbaugh 2017; Grano and Lasnik 2018).

matters most going forward is that low verb doubling lands lower in the clausal spine than the top of Group 1/2 complements. In the process we have committed ourselves to the maximal clause structure in (66). In the next section we flesh out our Williams Cycle treatment of improper verb doubling.

(66) *Clause structure and verb doubling in Mandarin Chinese*



4 Improper verb doubling and the Williams Cycle

We believe that improper verb doubling is best characterised as a special case of the Williams Cycle (Williams 1974, 2003, 2011, 2013), a locality constraint positing a quite general connection between the height of movement and the size of clausal complement it can escape. We flesh out a Level Embedding derivation of the Williams Cycle in Sect. 4.1, explicate how that derives the restrictions on verb doubling in Sect. 4.2, and verify a couple of predictions concerning multiple layers of clausal embedding in Sect. 4.3. In Sect. 4.4 we offer a solution to a non-intervention puzzle that arises from our analysis.

4.1 A Level Embedding Approach

We define the Williams Cycle as a descriptive constraint on movement of the form in (67). More or less descriptively restrictive versions of this constraint are special cases of the Williams Cycle. They all involve being more precise about the kinds of movement that are subject to this general Height-Locality constraint.

(67) *The Williams Cycle*

For certain types of movement, there is a principled connection between landing and locality such that: the higher movement lands in clausal functional sequence, the more unbounded it is.

There are a range of different theoretical models which derive some version of the Williams Cycle, all of which could explain improper verb doubling. For concreteness, we focus on the *Level Embedding* approach originally proposed by Williams (2003, 2011, 2013), and further developed by Poole (2022) and Meadows (2024). The major theoretical innovation of Level Embedding concerns the timing of main and embedded structure-building. The standard assumption is that we build an embedded clause before merging into the matrix clause, e.g., as an argument of the VP, and then go to finish off constructing the matrix clause. Level Embedding instead maintains that main and embedded clauses are built in parallel, in the sense of (68a). This is concretely illustrated in (68b). Whilst still in separate workspaces (notated by comma in subsequent diagrams), it is not possible to finish building one clause before another. Only embedding a clause can prevent it from being built to completion. Following Meadows (2024), we term this central assumption *Parallel Derivation*. This is the major theoretical ‘price’ to pay for use of Level Embedding architecture.

(68) *The major innovation of Level Embedding*

- a. PARALLEL DERIVATION: The same component of the clausal *fseq* (V, Voice, F, T, C, etc.) is externally merged in every workspace at the same time.
- b.

[CP Tom said [CP that he will cook the rice]]	
[VP say __], [VP cook the rice]	Merge V
[VoiceP Tom [VP say __]], [VoiceP he [VP cook the rice]]	Merge Voice
[TP Tom _i [VoiceP t _i [VP said __]]], [TP he _i will [VoiceP t _i [VP cook the rice]]]	Merge T
[CP [TP Tom _i [VoiceP t _i [VP said __]]], [CP that [TP he _i will [VoiceP t _i [VP cook the rice]]]]	Merge C

In combination with Parallel Derivation, the Level Embedding approach makes use of two devices which used in contemporary generative syntax to greater or less extents. The first device is *Late Merge* (Lebeaux 1988, 1998), for use

in clausal embedding. Something equivalent to Late Merge is escapable given a commitment to Parallel Derivation. Only the very smallest clauses can be embedded in the same cycle in which they are built. If we want to embed a CP anywhere other than at the CP-level (as an adjunct), we will have to wait around until both main and embedded clauses are CP-sized, and then Late Merge the CP into, e.g., an argument position of the VP.

(69) *Embedding CP complement with Late Merge*

[CP [TP Tom _i [VoiceP t _i [VP said ___]]]], [CP that [TP he _i will [VoiceP t _i [VP cook the rice]]]]	Build to CP
[CP [TP Tom _i [VoiceP t _i [VP said [CP that [TP he _i will [VoiceP t _i [VP cook the rice]]]]]]]]	Late Merge CP

The second device is a version of the *Strict Cycle Condition* (SCC) displayed in (70). Some version of the SCC is instantiated in the *Extension Condition* (Chomsky 1995), and *Featural Cyclicity* (Richards 1999).²⁰ The version of the SCC employs much more cycles than traditionally assumed, usually just CP and NP. For every projection in the clausal functional sequence, there is cycle of Merge/Agree associated with it. Once a new clausal F is merged into the spine, a new cycle starts and material introduced in prior cycles cannot be altered. The SCC essentially forbids Late applications of Merge, though we will stipulate that Late External Merge is unaffected. We return to tension between embedding with Late Merge and the Strict Cycle Condition in Sect. 6.3.

(70) *Strict Cycle Condition*

The output of prior cycles of rule application cannot be altered by rules in subsequent cycles. Every functional projection in the clausal *fseq* delineates a new cycle.

Long distance movement is restricted by the interaction between Parallel Derivation and the SCC. We take for granted that movement cannot leave a clause until it is embedded. By Parallel Derivation, the bigger a clause is, the later in the derivation it will be embedded. This waiting around for embedding is where the Strict Cycle comes in. Once the complement is embedded, subsequent long-movement is only possible to positions introduced in the same cycle in which embedding takes place. Any movement triggers introduced lower/earlier in the matrix clause (e.g., on T, F, etc.) have been forced to wait too long, as schematised below.

- | | | | |
|------|---|--|-----------------------------------|
| (71) | a. [VP] | | Parallel Derivation |
| | b. [CP [TP [VP]]], [CP [TP [VP]]] | | Parallel Derivation |
| | c. [CP [TP [VP [CP [TP [VP]]]]]] | | Embedding with Late Merge |
| | d. [CP ✓ [TP ✗ [VP ✗ [CP t [TP [VP]]]]]] | | SCC-compliant subsequent movement |

Combining a Strict Cycle Condition with Parallel Derivation and Late Merge derives a particularly strong version of the Williams Cycle. This is the *Generalised Ban on Improper Movement* (GBOIM) given in (72) (Williams 2003; Poole 2022). According the GBOIM, *all* movements must land at the same height or higher in the clausal *fseq*. We touch on whether this is overly restrictive in Sect. 6.3.

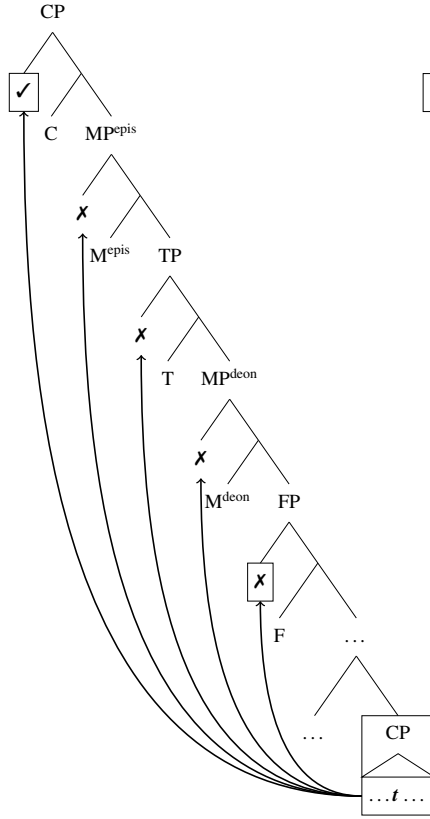
(72) *Generalised Ban on Improper Movement (GBOIM)*

Movement to [Spec, XP] cannot proceed from [Spec, YP] or across YP, where Y is higher than X in the clausal functional sequence (*fseq*). [Poole 2022:348, based on Williams 2003]

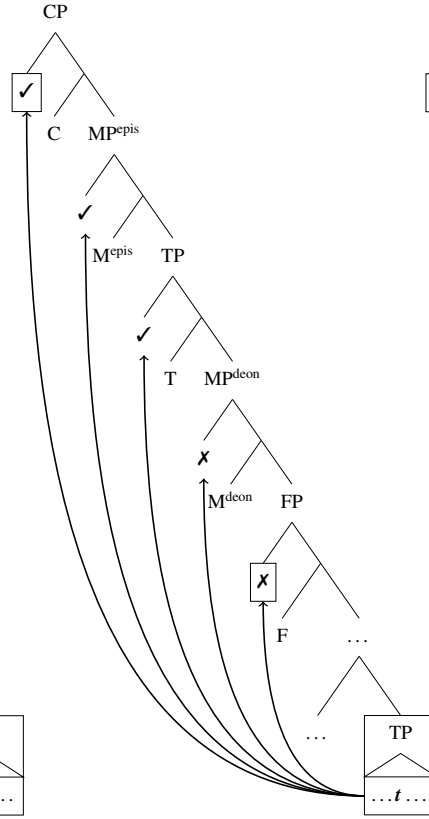
In the trees below, we display representations of movement constrained by the GBOIM out of CP-sized (73), TP-sized (74) and FP-sized (75) complements. Recall that these are three complement clause sizes motivated in Sect. 3.2. Movement out of a CP complement can land no lower than [Spec, CP] in the clausal *fseq* (73). This means that, crucially, VP-movement is not possible to the post-subject position we identified as [Spec, FP]. The same situation holds for movement from TP complements, where movement can land no lower than [Spec, TP] (74). Only in the case of FP complement, from which movement can land no lower than [Spec, FP] is VP-movement to that post-subject position possible (75).

²⁰ The Strict Cycle Condition does not enforce a total inaccessibility to syntactic operations enforced by the *Phase Impenetrability Condition* (Chomsky 2000, 2001). A probe can still find a goal in material from a previous cycle, e.g., a ϕ -probe on T finding a goal within the VP, and copy back the relevant features. This process is permitted because, crucially, it does not alter the goal. Since the probe itself is altered by Agree, Agree is constrained by the Strict Cycle both for its putative role in mediating movement, but also for processes of ϕ -agreement more generally. For a more in depth review of the SCC and its interactions with Merge, see Müller (2023) and Privzentseva (2023).

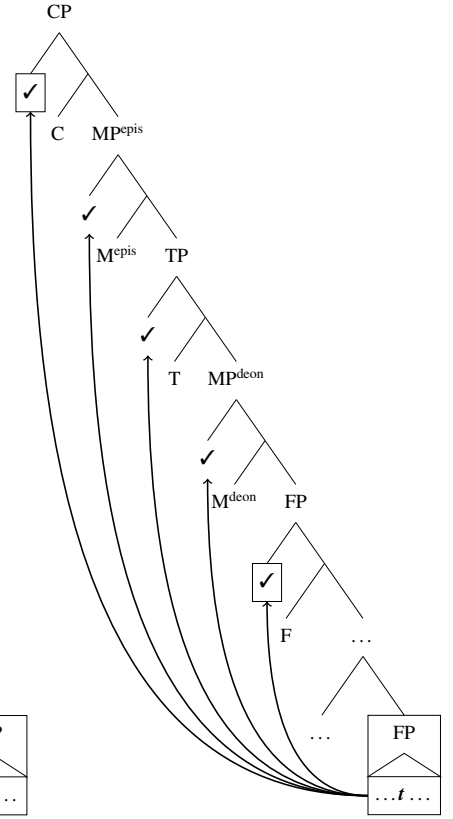
(73)



(74)



(75)



We have now seen the key ingredients of the Level Embedding approach, and how they derive a locality restriction of which improper verb doubling is a special case. We now go through the relevant derivations in a more mechanistic fashion, to make clear what precisely is a problem when.

4.2 Deriving improper verb doubling

Low VD out of Group 3 complements Recall that Group 3 clauses are complements to predicates like *chángshì* ‘try’. We analysed these complements as smaller than TP, since they lack distinct subjects and any modals. Nevertheless these complements can display clause-internal verb doubling. We therefore treat such complements as maximally FP-sized, i.e., built to the level at which low VP-movement takes places. Recall that it is only out of these complements that cross-clausal low verb doubling is possible. This follows from our assumptions so far.

We first build the main and complement clause to FP-size (76a)-(76b), according to Parallel Derivation. At this stage, the Cycle started by Merging F into the clausal spine, that we embed the complement into the matrix clause (76c). There is still time in the F-cycle for movement triggered at this cycle to occur. There is a featural trigger on matrix F, which attracts embedded VP to matrix [Spec, FP] (76d). This movement is completely compliant with the Strict Cycle Condition, since we affect only the material introduced in the present cycle. The FP-sized matrix clauses continues growing to the full CP-size, since there is nowhere to try and embed it (76e).

- (76) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [_{VP} try], [_{VP} read the novel]
- b. *Step 2: Build up to the FP-level* (Parallel Derivation)
 [_{FP} F_[uV] [_{VoiceP} Wassily try]], [_{FP} [_{VoiceP} read the novel quickly]]
- c. *Step 3: Embed the FP in the other FP* (Embedding with Late Merge)
 [_{FP} F_[uV] [_{VoiceP} Wassily try [_{FP} [_{VoiceP} read the novel quickly]]]]

- d. *Step 4: Probe on matrix F triggers VP-movement* (Strict Cycle ✓)
 [FP [VP read the novel] F_[#V] [VoiceP Wassily try [FP [VoiceP read DE quickly]]]]
- e. *Step 5: Build up to the CP-level in the matrix*
 [CP [TP Wassily [FP [VP read the novel] [VoiceP try [FP [VoiceP read DE quickly]]]]]]

Low VD out of Group 1 and 2 complements It is with other complement types that restrictions on cross-clausal verb doubling emerge. Remember that we analyse Group 1 complements as CP-sized, and Group 2 complements as TP-sized. Group 1 complements behave essentially like matrix clauses in their internal richness, whereas Group 2 complements stand somewhere between Group 2 and Group 3.

The crucial point is that both complements are built to higher in the clausal *fseq* than FP (as in (77a)-(77b) and (78a)-(78b)). It is this, combined with the Strict Cycle Condition, that produces a missed opportunity. Imagine that we build our main and to be embedded clauses to FP-level, where there is a featural trigger for VP-movement on matrix F. The problem is that unless we embed an FP complement, no such movement can take places. Suppose we wait until the clauses have grown to TP-sized for the Group 2 case (77c), or grown all the way to CP-sized for the Group 1 case (78c). In either situation we do late embedding, but movement to [Spec, FP] is forbidden by the Strict Cycle Condition (77d)/(78d).²¹

- (77) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [VP force Piet], [VP read the novel]
- b. *Step 2: Build up to the TP-level* (Parallel Derivation)
 [TP Wassily [FP F_[uV] [VoiceP force Piet]], [TP [FP [VoiceP read the novel quickly]]]]
- c. *Step 3: Embed the TP in the other TP* (Embedding with Late Merge)
 [TP Wassily [FP F_[uV] [VoiceP force Piet [TP [FP [VoiceP read the novel quickly]]]]]]
- d. *Step 4: Probe on matrix F cannot trigger VP-movement anymore* (Strict Cycle ✗)
 *[TP Wassily [FP [VP read the novel] F_[#V] [VoiceP force Piet [TP [FP [VoiceP read DE quickly]]]]]]
- (78) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [VP think], [VP read the novel]
- b. *Step 2: Build up to the CP-level* (Parallel Derivation)
 [CP [TP Wassily [FP F_[uV] [VoiceP think]]]], [CP [TP Piet [FP [VoiceP read the novel quickly]]]]
- c. *Step 3: Embed the CP in the other CP* (Embedding with Late Merge)
 [CP [TP Wassily [FP F_[uV] [VoiceP think [CP [TP Piet [FP [VoiceP read the novel quickly]]]]]]]]
- d. *Step 4: Probe on matrix F cannot trigger VP-movement anymore* (Strict Cycle ✗)
 *[CP [TP Wassily [FP [VP read the novel] F_[#V] [VoiceP think [CP [TP Piet [FP [VoiceP read DE quickly]]]]]]]]

To move to [Spec, FP] we can only do so at the F-cycle, precisely when embedding TP or CP complements is forbidden by Parallel Derivation. The derivation is thus forbidden in a pincer move by the competing requirements of Parallel Derivation and the Strict Cycle Condition.

High VD High verb doubling is unaffected by this pincer move of competing constraints, precisely because it lands as high in the functional sequence as biggest complement type is built to - the C/CP-level. In other words, there will be no complement embedding that happens in a later cycle than when high verb doubling occurs. As a result, movement to [Spec, CP] out of any size of complement complies with the Strict Cycle Condition. We simply build our main and embedded clauses in parallel (79a)-(79b), then embed at the relevant cycle (FP, TP or CP) (79c). Within the same, or later cycle (if embedding FP or TP), cycle movement of the embedded VP to [Spec, CP] is driven by a trigger on matrix C (79d)-(79e).

²¹ To be explicit, at the matrix F-cycle, the probe looks for a goal in its c-command domain and finds none; the relevant feature is located on a VP in the other workspace. Following Preminger (2014), we assume the obligatoriness associated with probes is the triggering of search, not in requiring a value.

- (79) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [_{VP} try], [_{VP} read the novel]
- b. *Step 2: Build up to the FP-level (or TP/CP-level)* (Parallel Derivation)
 [FP [_{VoiceP} Wassily try]], [FP [_{VoiceP} read the novel quickly]]
- c. *Step 3: Embed the FP in the other FP* (Embedding with Late Merge)
 [FP [_{VoiceP} Wassily try [FP [_{VoiceP} read the novel quickly]]]]
- d. *Step 4: Build up the CP-level in the matrix*
 [CP C_[uTop] [TP Wassily [FP [_{VoiceP} try [FP [_{VoiceP} read the novel quickly]]]]]]
- e. *Step 5: Probe on matrix C triggers VP-movement* (Strict Cycle ✓)
 [CP [_{VP} read the novel] C_[#Top] [TP Wassily [FP [_{VoiceP} try [FP [_{VoiceP} read DE quickly]]]]]]

4.3 Prediction: movement out of multiply-embedded clauses

In this section we show that VP-fronting over multiple levels of embedding conforms to the Williams Cycle that emerges from the Level Embedding approach.

4.3.1 Restricted verb doubling across multiple embeddings

The first type of multiple-embedding configuration is schematised in (80), where a non-Group 3 predicate is embedded under a Group 3. As already argued in Sect. 3.2, Group 3 predicates take FP-sized complements, while non-Group 3 ones take clauses bigger than FP as their complement. Since the VP in the most embedded clause needs to move out of an environment bigger than FP, we predict that low verb doubling targeting [Spec, FP] is restricted compared to its high counterpart landing in [Spec, CP] of the matrix.

- (80) *Non-Group 3 predicate embedded under Group 3*
 [CP ... [TP [FP ... [VoiceP Pred._{Group 3} [FP Pred._{Group 3} $\neg$ 3 [_{>FP} ... VP ...]]]]]]]]

Our prediction is first borne out by the configuration (81) where *xiāngxìn* ‘believe’, a Group 1 predicate, is embedded under *juédìng* ‘decide’, a Group 3 predicate. Since *xiāngxìn* takes a CP-sized clausal complement, the fronted VP is prohibited from landing in positions lower than CP. Therefore low verb doubling across multiple levels of embedding is not permitted in (81a), but high verb doubling in the same configuration is attested in (81b).

- (81) *Group 1 embedded under Group 3: no low VD*
- a. *Wassily **kàn nà-běn xiǎoshuō** *juédìng* [_{FP} *xiāngxìn* [_{CP} Piet *kàn de hěn kuài*]].
 Wassily read that-CL novel decide believe Piet read DE very quickly
 Int.: ‘Wassily decides to believe that Piet reads the novel quickly.’
- b. **Kàn nà-běn xiǎoshuō** Wassily *juédìng* [_{FP} *xiāngxìn* [_{CP} Piet *kàn de hěn kuài*]].
 read that-CL novel Wassily decide believe Piet read DE very quickly
 ‘As for reading the novel, Wassily decides to believe that Piet reads it quickly.’

We also correctly predict that the same contrast between low and high verb doubling exists in the configuration where a Group 2 predicate is embedded under a Group 3 one, as exemplified in (82). The Group 3 *chángshì* and the group 2 *qiǎngpò* respectively take FP- and TP-sized clausal complements. As the VP still has to move out of an embedded clause that is bigger than its low landing site, low verb doubling in (82a) is ruled out. By contrast, high verb doubling (82b) is licensed since the final landing site is higher than the TP complement containing the base-position.

- (82) *Group 2 embedded under Group 3: no low VD*
- a. *Wassily **kàn nà-běn xiǎoshuō chángshì** [_{FP} *qiǎngpò* Piet [_{TP} *kàn de hěn kuài*]].
 Wassily read that-CL novel try force Piet read DE very quickly
 Int.: ‘Wassily tries to force Piet to read the novel quickly.’

- b. **Kàn nà-běn xiǎoshuō** Wassily chángshì [_{FP} qiǎngpò Piet [_{TP} kàn de hěn kuài]].
 read that-CL novel Wassily try force Piet read DE very quickly
 ‘As for reading the novel, Wassily tries to force Piet to read it quickly.’

4.3.2 No restrictions on multiple Group 3 embeddings

We have seen that improper verb doubling occurs in configurations where the bigger-sized complement clauses are more embedded than the smaller-sized ones. The second prediction we make is if multiple-embedding configurations consist of Group 3 predicates only (as schematised in (83)), then neither low nor high verb doubling is restricted.

- (83) *Group 3 predicate embedded under Group 3*
 [CP ... [TP [FP ... [VoiceP Pred._{Group 3} [FP Pred._{Group 3} [FP ... **VP** ...]]]]]]]

This prediction is corroborated empirically by (84), in which the Group 3 predicate *juédìng* embeds another Group 3 *chángshì*. Since both of them take FP-sized clausal complements, moving the VP from the most embedded clause to the matrix [Spec, FP] does not cross any higher projections in the clausal *fseq*. Thus low verb doubling in such a configuration is allowed, as shown in (84a). High verb doubling is also unaffected as in (84b).

- (84) *Group 3 embedded under Group 3: no restrictions*
- a. Wassily **kàn nà-běn xiǎoshuō** juédìng [_{FP} chángshì [_{FP} kàn de hěn kuài]].
 Wassily read that-CL novel decide try read DE very quickly
 ‘Wassily decides to try to read the novel quickly.’
- b. **Kàn nà-běn xiǎoshuō** Wassily juédìng [_{FP} chángshì [_{FP} kàn de hěn kuài]].
 read that-CL novel Wassily decide try read DE very quickly
 ‘As for reading the novel, Wassily decides to try to read it quickly.’

There is no constrain on how many Group 3 predicates can be stacked together. Our further prediction is that both low and high verb doubling are still permitted even if the multiple-embedding configuration is made up of more than two Group 3 predicates. This is validated by the configuration in (85) where three Group 3 predicates are embedded consecutively one after another.

- (85) *Triple Group 3 embedding: no restrictions*
- a. Wassily **kàn nà-běn xiǎoshuō** juédìng [_{FP} kāishǐ [_{FP} chángshì [_{FP} kàn de hěn kuài]]].
 Wassily read that-CL novel decide begin try read DE very quickly
 ‘Wassily decides to begin trying to read the novel quickly.’
- b. **Kàn nà-běn xiǎoshuō** Wassily juédìng [_{FP} kāishǐ [_{FP} chángshì [_{FP} kàn de hěn kuài]]].
 read that-CL novel Wassily decide begin try read DE very quickly
 ‘As for reading the novel, Wassily decides to begin trying to read it quickly.’

Having confirmed these basic predictions of the Level Embedding Approach, we conclude with a puzzle concerning the featural status of low VD which we have abstracted away from thus far.

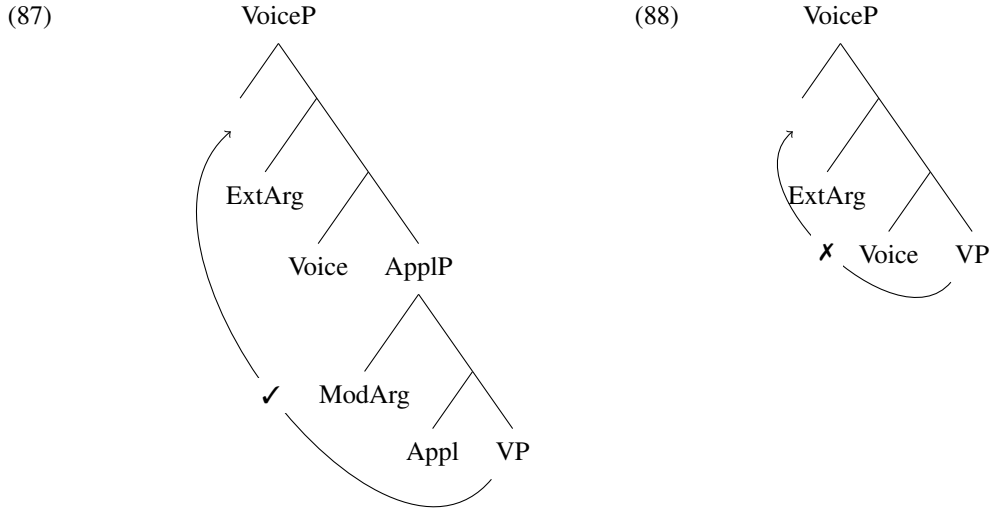
4.4 Solving a non-intervention puzzle

In Sect. 2.2 we proposed that low verb doubling is triggered by a categorial feature [*uV*] on F, which is often used for obligatory/EPP-like VP-movement (e.g., Travis and Massam 2021; van Urk 2024), because low verb doubling is not informationally marked (recall (11) in Sect. 2.1). Without further assumptions, a [*uV*] probe will encounter the most local (and therefore preferred) [V] goal that is part of the embedding predicate. By default then, our featural commitment leads to low verb doubling being clause-bound. Empirically this is problematic for Group 3 clausal complements, which seem to be transparent for low verb doubling. It is also problematic for central contention of this paper, since the feature-induced minimality rids us of the need to appeal to the Williams Cycle/Level Embedding to explain clause-boundedness.

We clearly need to address how it is possible in principle for $[uV]$ to ‘ignore’ intervening embedding material. Our intuition is that by default no embedding predicate constitutes a true intervener for $[uV]$ on F.²² Although the most local goal for Agree, movement of VPs in this configuration is independently ruled out. Agree works in such a way that it can apply again, after this failure, leaving the embedded VP the closest remaining goal.²³ The intended derivation is sketched below in (86). We are representing a stage of the derivation at the FP-cycle, after we have embedded an FP-sized complement. We build on Lai (2021, 2024) for the source of restriction on moving embedding VPs.

- (86) a. $\begin{array}{c} [\text{FP } F_{[uV]} [\text{VoiceP } \text{VP } [\text{FP } [\text{VoiceP } \text{VP }]]] \\ \text{O} \quad \text{AGREE } \checkmark \quad \text{O} \\ \text{---} \end{array}$ Agree with matrix VP $\checkmark$
- b. $\begin{array}{c} [\text{FP } F_{[uV]} [\text{VoiceP } \text{VP } [\text{FP } [\text{VoiceP } \text{VP }]]] \\ \text{O} \quad \text{MOVE } \times \quad \text{O} \\ \text{---} \end{array}$ Attract matrix VP to [Spec, FP] $\times$
- c. $\begin{array}{c} [\text{FP } F_{[uV]} [\text{VoiceP } \text{VP } [\text{FP } [\text{VoiceP } \text{VP }]]] \\ \text{O} \quad \text{AGREE } \checkmark \quad \text{O} \\ \text{---} \end{array}$ Agree with embedded VP $\checkmark$
- d. $\begin{array}{c} [\text{FP } F_{[uV]} [\text{VoiceP } \text{VP } [\text{FP } [\text{VoiceP } \text{VP }]]] \\ \text{O} \quad \text{MOVE } \checkmark \quad \text{O} \\ \text{---} \end{array}$ Attract embedded VP to [Spec, FP] $\checkmark$

There is a tight descriptive connection between VP-modification and the presence of verb doubling, as we noted in Sect. 2. Lai (2021, 2024) proposes that postverbal modifiers are introduced as arguments of an intermediate projection between VoiceP/vP, the external argument (ExtArg) introducer, and the VP (87). This intermediate, applicative-like projection is not present in the construction of VPs without this additional modifier argument (ModArg) (88). In none of our examples is there an embedding predicate modified in this way. By hypothesis these embedding thematic domains are of the more truncated variety in (88). It is this truncated structure, alongside a further locality constraint, that restricts embedding VP-movement.



²² We are aware of another possible solution: Group 3 predicates are actually functional restructuring verbs (Cinque 2004; Wurmbrand 2004), lacking [V], and thus do not count as interveners. This would require us to reanalyse Group 3 complements as monoclausal structures, and it would permit a straightforward featural intervention account of low verb doubling’s clause-boundedness without needing the Williams Cycle. If this solution turns out to be preferable to what we pursue in the main text, we could still adopt it. It is the *combination* of improper verb doubling and restrictions on long-distance object shift (in Sect. 5) which particularly favour the Williams Cycle. We would merely concede that improper verb doubling alone can be handled equally well by the intervention approach. Nevertheless we are not certain that Group 3 predicates really resemble *xiǎngyào* ‘want’, which Grano (2015) argues to be a functional restructuring predicate. For this reason we prioritise our Anti-locality solution in the main text.

²³ Here could appeal to either *Multiple Agree* (Hiraiwa 2001, 2005) or *Interaction and Satisfaction Agree* (Deal 2015, 2024). In the latter case we would need to say that $[uV]$ on F has no Satisfaction Condition, allowing it to interact with any [V] in its c-command domain at the FP-cycle. The combination of the Strict Cycle Condition and Embedding with Late Merge restrict looking past the embedding V into the complement clauses for complements bigger than FP.

The locality constraint we need, following Lai (2021, 2024), is *Comp-to-Spec Anti-locality* (Abels 2003), as given in (89). Suppose that Voice, the external argument introducer, is a phase head and thus movement across it must land in its specifier (further assuming that multiple specifiers are possible). The problem is that a VP which is the complement to Voice, such as in (88), is ‘too close’ to be re-merged at [Spec, VoiceP]. The extra layer of structure between Voice and VP present with suitable modifiers, for instance in (87), gives sufficient distance to allow VP-movement.

- (89) *Comp-to-Spec Anti-locality Constraint*
 Movement of the complement of a head X must cross a head other than X. [Abels 2003:12-14, *et seq.*]

All unmodified embedding predicates, by virtue of their truncated VP structures, do not allow the matrix VP to move to that crucial clause-internal edge. This is why matrix VPs do not generally count as interveners. [*uV*] on matrix F can only subsequently find a suitable goal in FP-sized complements, since movement of them will not violate the Strict Cycle Condition.

Now that we have addressed this lacuna in our analysis, we wrap up this section. We outline a version of Level Embedding, a theoretical architecture which derives a fairly strict version of the Williams Cycle. We have seen how this derives improper verb doubling effects. In the next section we turn our attention to long-distance object shift. We see clear evidence for the Williams Cycle in Height-Locality effects extending across movement of *different categories and featural triggers*.

5 Clause-bound movement of nominals: object shift

In Sect. 3.2.1, we have seen that object shift does not behave identically in various embedding contexts. We have used the distinct distribution of object shift as a crucial argument to favour our proposal that embedding predicates take clausal complements with different sizes. Unlike long-distance VP-fronting discussed throughout the paper, long-distance object shift has received considerable attention in the literature, such as Ernst and Wang (1995), Shyu (1995), Paul (2002b, 2005), Chen (2023, to appear), among many others. The purpose of the current section is to show how our approach can be extended to those cases.

5.1 Outlining the basic paradigm

We first present the full paradigm of object shift in different embedding contexts in what follows. We construct the examples in exactly the same environments as long-distance verb doubling in Sect. 3.1. This ensures that the distribution of long-distance object shift is only affected by the size of complement clause from which the nominal movement starts.

Within Group 1 complement clauses, we have already known that object shift can target two positions due to their CP-sized status. One of them follows the embedded subject, and another precedes the same subject, as respectively given again in (90a) and (90b). Similar to the way of distinguishing verb doubling, we dub the former situation *low object shift* and the latter *high object shift* from this point onwards. When the object moves across the full clause-boundary, low and high object shift show the same dichotomy as their verb doubling counterparts: shifting the object to the lower site of the matrix is banned (90c), whereas landing in the higher site is permitted (90d). We treat (90c) as an instance of *improper object shift*.

- (90) a. Wassily rènwéi [Piet **nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily think Piet that-CL novel read DE very quickly
 ‘Wassily thinks that Piet reads THAT NOVEL quickly.’
 ‘think’ + embedded low OS ✓ (=31a)
- b. Wassily rènwéi [**nà-běn xiǎoshuō** Piet kàn de hěn kuài].
 Wassily think that-CL novel Piet read DE very quickly
 ‘Wassily thinks that, THAT NOVEL, Piet reads quickly.’
 ‘think’ + embedded high OS ✓ (=31b)
- c. *Wassily **nà-běn xiǎoshuō** rènwéi [Piet kàn de hěn kuài].
 Wassily that-CL novel think Piet read DE very quickly

Int.: ‘Wassily thinks that Piet reads THAT NOVEL quickly.’

‘think’ + matrix low OS ✗

- d. **Nà-běn xiǎoshuō** Wassily rènwéi [Piet kàn de hěn kuài].
that-CL novel Wassily think Piet read DE very quickly
‘As for THAT NOVEL, Wassily thinks Piet reads it quickly.’

‘think’ + matrix high OS ✓

We have also seen that embedded object shift is possible in the Group 2 embedding context (91a), and argued that Group 2 clausal complements are TP-sized due to the absence of overt embedded subject and sentence-level adverb (recall Sect. 3.2.1). What is intriguing here is that the potential occurrence of improper object shift is not attested. As shown in (91b), long-distance object shift is allowed in the matrix clause-medial position, unlike its low verb doubling counterpart in the same environment (27b). Moving the object to the matrix clause-initial position is permitted as usual.

- (91) a. Wassily qiǎngpò Piet [**nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily force Piet that-CL novel read DE very quickly
‘Wassily forces Piet to read THAT NOVEL quickly.’

‘force’ + embedded OS ✓ (=33)

- b. Wassily **nà-běn xiǎoshuō** qiǎngpò Piet [kàn de hěn kuài].
Wassily that-CL novel force Piet read DE very quickly
‘Wassily forces Piet to read THAT NOVEL quickly.’

‘force’ + matrix low OS ✓

- c. **Nà-běn xiǎoshuō** Wassily qiǎngpò Piet [kàn de hěn kuài].
that-CL novel Wassily force Piet read DE very quickly
‘As for THAT NOVEL, Wassily forces Piet to read it quickly.’

‘force’ + matrix high OS ✓

Proceeding to object shift in the Group 3 embedding context, we have been aware that embedded object shift in such an environment is not possible (as in (92a)) because of its most reduced, FP-sized status. However, no restrictions are imposed on the movement of object across Group 3 complement clauses. As given in (92b) and (92c), the shifted object can flexibly target its landing site either below or above the surface subject.

- (92) a. *Wassily chángshì [**nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily try read that-CL novel DE very quickly
Int.: ‘Wassily tries to read THAT NOVEL quickly.’

‘try’ + embedded OS ✗ (=35)

- b. Wassily **nà-běn xiǎoshuō** chángshì [kàn de hěn kuài].
Wassily that-CL novel try read DE very quickly
‘Wassily tries to read THAT NOVEL quickly.’

‘try’ + matrix low OS ✓

- c. **Nà-běn xiǎoshuō** Wassily chángshì [kàn de hěn kuài].
that-CL novel Wassily try read DE very quickly
‘As for THAT NOVEL, Wassily tries to read it quickly.’

‘try’ + matrix high OS ✓

The full paradigm of object shift in three embedding contexts is summarised in (93). The variations in embedded object shift have already been explained by the size of different clausal complements in Sect. 3.2.1. The key problem is the different distributions of long-distance object shift in the matrix clause-medial position. Object shift to the matrix left periphery is always clause-unbounded, as with matrix high verb doubling (cf. (25)). For the sake of space we will not discuss these cases in what follows, assuming that the Level Embedding approach to long-distance high verb doubling outlined in Sect. 4.2 fully applies to long-distance high object shift as well.

- (93) *Full paradigm of object shift in different embedding contexts*

Embedding predicate group	Matrix high OS	Matrix low OS	Embedded OS
Group 1: ‘think’ + [CP ...]	✓	✗	✓
Group 2: ‘force’ + [TP ...]	✓	✓	✓
Group 3: ‘try’ + [FP ...]	✓	✓	✗

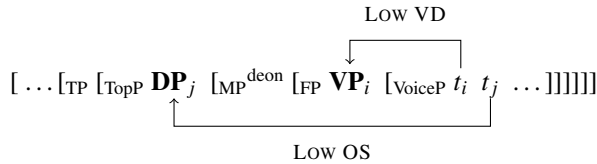
5.2 The landing site of low object shift vs. low verb doubling

Before presenting our analyses of long-distance low object shift, we need to highlight an important difference between low object shift and low verb doubling concerning their landing sites after movement. Recall that at the end of Sect. 3.2.1, we reached the conclusion that low verb doubling lands into a position lower than low object shift in the clausal spine. This is motivated by the empirical observation that within Group 3 complement clauses, verb doubling is possible while object shift is not. The contrast is illustrated again in (94).

- (94) a. Wassily chángshì [FP **kàn nà-běn xiǎoshuō** *kàn de hěn kuài*].
Wassily try read that-CL novel read DE very quickly
‘Wassily tries to read the novel quickly.’ Embedded VD in Group 3 ✓ (=28a)
- b. *Wassily chángshì [FP **nà-běn xiǎoshuō** *kàn de hěn kuài*].
Wassily try read that-CL novel DE very quickly
Int.: ‘Wassily tries to read THAT NOVEL quickly.’ Embedded OS in Group 3 ✗ (=35/92a)

That (94a) is well-established lays the foundation of our proposal that Group 3 complement clauses are FP-sized, since verb doubling can target the clause-medial position [Spec, FP] and occur in such embedded configurations. Then the ungrammatical (94b) gets explained right away once we adopt the assumption that object shift generally targets a position higher than the clause size of Group 3 complements; otherwise, we would predict no difference between (94a) and (94b). Following Paul (2002b, 2005) and Chen (2023, to appear), we assume that there is a TopP in the TP-domain, and that low object shift is triggered by [*u*Top] on this Top head and moves to its specifier position. A more articulated clause-medial structure is illustrated in (95), with both low verb doubling and low object shift indicated (cf. 66).

- (95) *Clause-medial structure and low object shift & verb doubling in Mandarin Chinese*



The different landing sites of low object shift and low verb doubling lead us to expect that their movement behaviour is not totally identical. As pointed out in the previous section, in the Group 2 embedding context, low object shift does not manifest improper movement (e.g., (91b)), in contrast to low verb doubling that does behave improperly (e.g., (27b)). In the following section we show how the Level Embedding approach can capture this contrast as well as other instances of proper/improper object shift attested in the clause-medial position..

5.3 Deriving long-distance low object shift

Low OS out of Group 1 complements We begin with the only situation where improper object shift is observed, namely low object shift out of Group 1 complements, as exemplified in (90b). According to Parallel Derivation, we start building up the main and embedded clause simultaneously from VP (96a) to CP (96b). Once reaching the CP-cycle, we embed the CP complement into the matrix CP via Late Merge (96c). However, what is forbidden in the CP-cycle is to move the object from the embedded CP complement to the matrix [Spec, TopP], since such a movement violates the Strict Cycle Condition (96d), thereby giving rise to improper object shift.

- (96) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
[_{VP} think], [_{VP} read the novel]

- b. *Step 2: Build up to the CP-level* (Parallel Derivation)
[CP [TP Wassily [TopP Top_[uTop] [VoiceP think]]], [CP [TP Piet [TopP [VoiceP read the novel quickly]]]]
- c. *Step 3: Embed the CP in the other CP* (Embedding with Late Merge)
[CP [TP Wassily [TopP [VoiceP think [CP [TP Piet [TopP [VoiceP read the novel quickly]]]]]]]]
- d. *Step 4: Probe on matrix Top cannot trigger DP-movement anymore* (Strict Cycle ✗)
*[CP [TP Wassily [TopP [DP the novel] Top_[uTop] [VoiceP think [CP [TP Piet [TopP [VoiceP read DE quickly]]]]]]]]

Low OS out of Group 2 complements The derivation of long-distance low object shift out of Group 2 complements also starts with building up both clauses to the VP-level in parallel (97a). Parallel Derivation keeps going to the TopP-level (97b). In this cycle, the TopP-sized complement clause is embedded (using Late Merge) into the congruent matrix clause (97c), and the feature on the matrix Top head triggers the nominal movement to its specifier (97d). Once object shift is done, the matrix clause continues to grow and ends up at the CP-level (97e).

- (97) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
[VP force Piet], [VP read the novel]
- b. *Step 2: Build up to the TopP-level* (Parallel Derivation)
[TopP Top_[uTop] [VoiceP Wassily force Piet]], [TopP [VoiceP read the novel quickly]]
- c. *Step 3: Embed the TopP in the other TopP* (Embedding with Late Merge)
[TopP Top_[uTop] [VoiceP Wassily force Piet [TopP [VoiceP read the novel quickly]]]]
- d. *Step 4: Probe on matrix Top triggers DP-movement* (Strict Cycle ✓)
[TopP [DP the novel] Top_[uTop] [VoiceP Wassily force Piet [TopP [VoiceP read DE quickly]]]]
- e. *Step 5: Build up to the CP-level in the matrix*
[CP [TP Wassily [TopP [DP the novel] [VoiceP force Piet [TopP [VoiceP read DE quickly]]]]]]

The derivation above shows that the clause size of Group 2 complements can actually be smaller than TP (i.e., TopP) because of the Strict Cycle Condition. As mentioned in fn. 2, we use TP (and also CP) as an umbrella term throughout the paper. That is to say, what we call TP in principle can split into several functional projections, e.g., the articulated clause-medial structure given in (95). Adopting this split-TP assumption would not undermine what we have said previously about improper verb doubling in the Group 2 embedding context (recall Sect. 4.2): as long as the size of Group 2 complement clauses is bigger than FP in the clausal *fseq*, low verb doubling is not permitted to land in the matrix [Spec, FP]. Abstracting away from the precise size of Group 2 complements, we will stick to the label TP till the end of the paper.

Low OS out of Group 3 complements Low object shift out of Group 3 complements does not significantly differ from its Group 2 counterpart. Again, we build up two clause structures in parallel from VP (98a) to FP (98b). Clause embedding with Late Merge happens in this FP-cycle (98c), followed by the matrix clause being built up to the Top-level (98d). The feature on the matrix Top triggers low object shift at this level without violating the Strict Cycle Condition (98e). In the end the matrix clause grows up and stops at the CP-level (98f).

- (98) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
[VP try], [VP read the novel]
- b. *Step 2: Build up to the FP-level* (Parallel Derivation)
[FP [VoiceP Wassily try]], [FP [VoiceP read the novel quickly]]
- c. *Step 3: Embed the FP in the other FP* (Embedding with Late Merge)
[FP [VoiceP Wassily try [FP [VoiceP read the novel quickly]]]]
- d. *Step 4: Build up to the TopP-level in the matrix*
[TopP Top_[uTop] [FP [VoiceP Wassily try [FP [VoiceP read the novel quickly]]]]]]
- e. *Step 5: Probe on matrix Top triggers DP-movement* (Strict Cycle ✓)
[TopP [DP the novel] Top_[uTop] [FP [VoiceP Wassily try [FP [VoiceP read DE quickly]]]]]]

- f. *Step 6: Build up to the CP-level in the matrix*
 [CP [TP Wassily [TopP [DP the novel] [FP [VoiceP try [FP [VoiceP read DE quickly]]]]]]]

5.4 Three locality profiles for embedding contexts

By comparing the behaviour of cross-clausal verb doubling and cross-clausal object shift, we can construct locality profiles for our three embedding contexts - repeated from the introduction in (99). Complements to Group 1 predicates are opaque to the most kinds of movements (low verb doubling and low object shift). At the other end of the scale, complements to Group 3 predicates are transparent to everything. Complements to Group 2 predicates occupy an intermediate status in being opaque to just low verb doubling.²⁴

(99)	Locality profile	Pred. _{Group 1} + [CP ...]	Pred. _{Group 2} + [TP ...]	Pred. _{Group 3} + [FP ...]
	Transparent to:	High VD/OS	High VD/OS, Low OS	High VD/OS, Low VD/OS
	Opaque to:	Low VD/OS	Low VD	∅

Showing that two categorially distinct kinds of movement strengthens the case for treating these Height-Locality effects as instances of the Williams Cycle. Looking at each movement in isolation, it is possible in principle to imagine probe-specific restrictions which do not concern their height in the clausal spine.²⁵ However, the more we find different types of movement within a language that display *prima facie* Height-Locality effects, the probability of a merely coincidental connection to position decreases. The Level Embedding approach to the Williams Cycle expects Height-Locality effects to hold of movements with different featural triggers.

6 Theoretical issues arising

This section concludes the paper by reviewing some of the theoretical issues which arise from treating improper verb doubling and restrictions on object shift as special cases of the Williams Cycle. We touch on the viability of the A/ $\bar{A}$ locality distinction in Sect. 6.1, why something like the Williams Cycle cannot be easily avoided in Sect. 6.2, and some of the weaknesses of the Level Embedding approach in Sect. 6.3.

6.1 Locality beyond the A/ $\bar{A}$ binary

Low verb doubling clearly does not behave like archetypal A-movement, since other VPs do not count as interveners (recall Sect. 4.4). This is evident in the very possibility of long distance low verb doubling out of Group 3 complements. Nevertheless we cannot simply classify low verb doubling as $\bar{A}$ -movement, on account of its obvious clause-boundedness. One could characterise low verb doubling as a kind of mixed A/ $\bar{A}$ dependency, at least as a descriptive short-hand. This cannot stand in for a proper analysis of locality profile we observe.

The literature on mixed A/ $\bar{A}$ dependencies generally focuses on movement types with mixture of locality, morphosyntactic and interpretative properties from traditional A and $\bar{A}$ classes (van Urk 2015; Longenbaugh 2017; Scott 2021; Lohninger et al. 2022; Chen 2023; Lohninger and Yip 2023; Chen to appear, a.o.). Movement that, for example, is possible long distance, feeds case/ ϕ -agreement and can create new binding dependencies. The widespread assumption popularised by van Urk (2015) is that such movements are driven by a combination of featural triggers, which are individually associated with the A or $\bar{A}$ profile of properties and together add up to combined profile.

This composite-probe logic, which builds ternary distinction out of a binary one, is not optimised for dealing with ‘mixed locality’. Low verb doubling, and a number of Height-Locality cases, display a locality profile which is characteristic of *neither A nor $\bar{A}$ movement, rather than both at once*. If locality phenomena still fall through cracks of a ternary A, $\bar{A}$ and A/ $\bar{A}$ -distinction, we suggest that this is because the underlying binary distinction is not fit for purpose. This conclusion was already reached on a more limited empirical picture by Williams (2003).

²⁴ There are of course other types of movement which could be used to enrich this picture further. We note that object *wh*-phrases can appear in pre-subject or post-subject position as well. Such *wh*-objects are restricted from post-subject position under very similar circumstances like regular shifted objects, i.e., not if they relate to a gap in a Group 1 complement. This may follow from the lack of *wh*-movement process that is distinct from topicalisation in Mandarin Chinese (Pan 2011).

²⁵ As we noted in fn. 22, one could entertain the possibility of the clause-boundedness of low verb doubling arising from A-like minimality considerations. This depends on assumptions about Group 3 predicates which may not be plausible.

The Williams Cycle suggests that there are at least as many types of movement, in terms of locality, as there are distinct landing sites for movement. Further types emerge for multiple possible featural triggers for a given position, each type of trigger with its own kind of minimality effects. The A/ $\bar{A}$ -distinction is simply a special case of this more general picture. The division of labour between positions and features in accounting for the rest of the A/ $\bar{A}$ -distinction (e.g., reconstruction, binding, weak crossover, interactions with case/agreement, etc.) remains an open question.

6.2 No escape from the Position Intuition

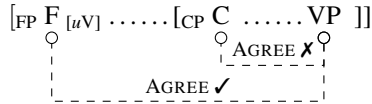
We have seen that Height-Locality effects can be profitably analysed by using the Williams Cycle, and in particular the Level Embedding approach. The Williams Cycle is not in fact the only possible account of this phenomenon. It belongs to a larger class of proposals, including the BOIM (e.g., Chomsky 1973, 1981; May 1979) and the *Universal Constraint on Operational Ordering* (Abels 2007, 2009), which all (implicitly) assume (100). This Position Intuition can be contrasted with a *Content Intuition*, under which locality is determined (purely) by the featural/semantic content of the movement trigger.²⁶ Avoiding the Williams Cycle is possible, but there is no escape from the Position Intuition.

(100) *The Position Intuition*

The locality of movement depends its location in a hierarchy of positions, or an order of operations.

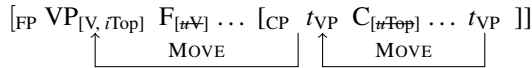
This is clearest if one were to take a superficially more conservative analysis of clause-boundedness effects. There is strand of literature which pursues the hypothesis that clause-boundedness involves in the inability to access certain intermediate positions, without which Phase Theory rules out the ‘one fell swoop’ attempts at movement (e.g., Longenbaugh 2017; Grano and Lasnik 2018; Chen 2023). This literature assumes that it is a lexical quirk of some functional categories to lack featural triggers. Thus for example, low verb doubling would be impossible out of embedded CPs because it is driven by featural trigger which is present on F but not embedded C, schematised below in (101). Since the embedded VP cannot be attracted to [Spec, CP], for want of a suitable probe, is not accessible to the suitable final probe, even though that last movement step is not inherently problematic.

(101) *The missing trigger approach to clause-boundedness*



A lot rests here on determining what a ‘suitable intermediate probe’ is. High verb doubling is possible out of embedded CP, and it must therefore be assumed that it is driven by suitable intermediate probe on embedded C. This raises an obvious question: why can a VP not undergo movement to [Spec, CP] triggered by the high VD probe, then undergo another step of movement to [Spec, FP] triggered by the low VD probe? The derivation in question is schematised below in (102). It will not be satisfactory to rule out chains composed of multiple movement types, since these are after all widely observed (e.g., raising plus *wh*-movement). What seems to matter is the *order* of probing different features on a single target, if intermediate movement does indeed play a key role in clause-boundedness.²⁷

(102) *A forbidden Mix-and-Match derivation*



In the worst possible case, one has to resort at this point to constraint on probe ordering which entirely recapitulates the BOIM, e.g., something that has been a target for an $\bar{A}$ -probe, cannot subsequently be targeted by an A-probe.²⁸ A ‘Ban on Improper Probing’ is just another implementation of the Position Intuition, and does not follow from principles enforcing minimality or successive-cyclic movement any more than the Williams Cycle does. The appeal of

²⁶ These intuitions are not necessarily in tension with each other. It seems plausible to us that there are restrictions that follow from Position (Height-Locality effects) and those that follow from Content (Intervention effects).

²⁷ The Williams Cycle is compatible with intermediate movement, at least through [Spec, CP], but does not rely such movement for its efficacy.

²⁸ The value bought in replacing A and $\bar{A}$ -positions with features, i.e., accounting for mixed positions, essentially evaporates here. One will have to stipulate where compound A+ $\bar{A}$ probes fall in this hierarchy in ways that will not seem obvious from their constituent parts.

the Williams Cycle over other implementations of the Position Intuition is precisely that it relies on an ordering that we cannot escape from: the clausal functional sequence.

6.3 The limitations of the Level Embedding approach

Our commitment in this paper is ultimately to some version of the Williams Cycle. We chose to exemplify the Level Embedding approach for concreteness, but as far as we can see other approaches like *Horizons* (Keine 2016, 2019, 2020) would be equally effective. Here we briefly dwell on a couple of the drawbacks of the Level Embedding approach that might motivate an alternative implementation.

At an empirical level the major challenge for Level Embedding concerns exceptions to the GBOIM-locality. It does seem that there are movements which can land lower in the clausal *fseq*. Two particularly problematic cases are: (i) intermediate movement through [Spec, ν P] out of CP-sized clause (e.g., quantifier stranding at ν P in West Ulster English (103a); see van Urk 2020 for a review), and (ii) varieties of raising that can land at [Spec, TP] or [Spec, ν P] having crossed a CP boundary (e.g., hyperraising to subject in Brazilian Portuguese (103b); see Zyman 2023 for a review). Unless we are willing to reanalyse all such cases, it seems like we have to carve out exceptions to the GBOIM. It remains an open question whether Level Embedding can admit such exceptions in a principled way.

- (103) a. [CP **What**_{*j*} did he [ν P [**all** *t_j*]_{*i*} say [CP that he did *t_i* on holiday]]]? [Henry 2012:ex. 26]
- b. [CP [TP **Os meninos**_{*i*} parecem [CP que *t_i* fizeram a tarefa]]].
the boys seem.3PL that did.3PL the homework
 ‘The boys seem to have done their homework.’ [Nunes 2009:ex. 2]

There is similarly thorny theory internal problem. The Level Embedding approach works by making movements to certain positions violations of the Strict Cycle Condition, and yet the method for doing so (Embedding with Late Merge) itself involves external Merge that by all rights should violate the Strict Cycle Condition too. The Level Embedding approach can only avoid this contradiction by exempting clausal embedding from the Strict Cycle Condition. As a brute stipulation this is clearly unattractive. If some instances of Merge are exempt from the Strict Cycle Condition, why not others?

We do not believe that these problems are insurmountable. Meadows (2024), for instance, argues that the empirical and theoretical problems of Level Embedding are two sides of the same coin: the need to effectively weaken the Strict Cycle Condition.²⁹ His suggestion, which does not get fully developed, is that successive-cyclic movement, hyperraising and clausal-embedding all involve Merge that is not feature-driven. All these instances of Merge are driven to make syntactic structure legible to PF (in terms of linearisation) or LF (in terms of semantic argument satisfaction). If the Strict Cycle Condition only pays attention to changes related to feature-checking, then any purely interface-driven movement will be able to apply to material introduced in earlier cycles. We leave developing such a strategy for future research.

Having addressed these outstanding theoretical issues, we can now conclude the paper. We have presented a detailed case study of VP and DP-movement showing Height-Locality effects. We believe that the best analysis of these effects appeals to some version of the Williams Cycle. Beside this general theoretical interest, our analysis enriches understanding about the clause-structure of Mandarin Chinese. That there are three sizes of complement clause, for example, would not be obvious on a cursory examination, especially given how Mandarin lacks many traditional morphosyntactic diagnostics of finiteness. Close attention to the locality and landing sites of movement remains a productive way to judge the analytically-necessary richness of clause structure.

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²⁹ The strategy pursued Williams (2003) and Poole (2022), is simply to reanalyse the problematic cases of movement as GBOIM compliant, and to propose that clausal-embedding involves an operation qualitatively different from Merge that by its very nature circumvents the Strict Cycle Condition. We are more inclined to modify the Level Embedding approach than to defend its current form to the hilt.

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